



Psychology in Germany

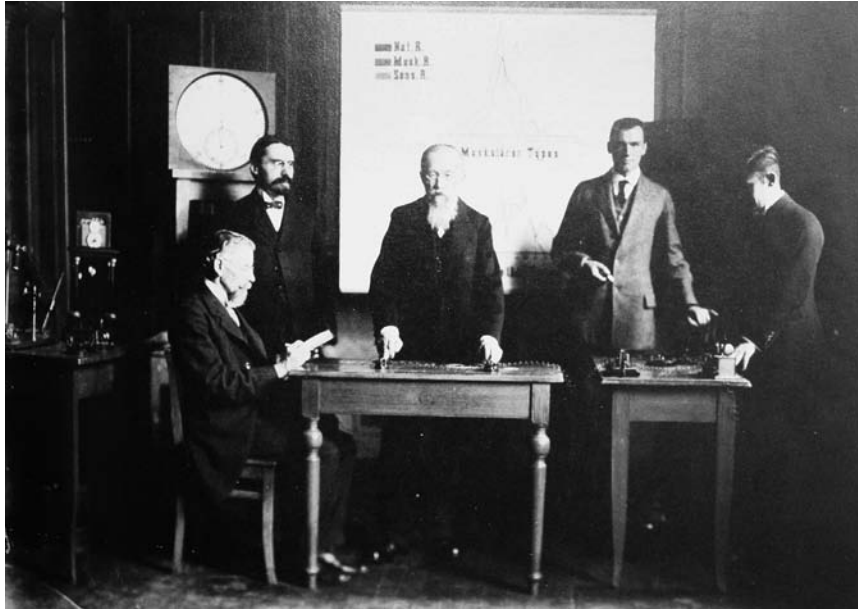
THE ACADEMIC DISCIPLINE OF SCIENTIFIC PSYCHOLOGY WAS FOUNDED institutionally in Germany at the end of the 19th century. It was a natural outgrowth of the progressive German university system, which was hospitable to the development of new disciplines such as linguistics and psychology. Wilhelm Wundt (1832–1920), who founded scientific psychology in Germany in 1879, the year he set up his experimental laboratory at the University of Leipzig, characterized the new discipline as *physiological psychology*. This was not because he believed that psychological states and processes must be reductively explained in terms of physiological states and processes, but because he believed that scientific psychology should appropriate the experimental methods that had proved so successful in the development of 19th-century German physiology.

Wundt's new experimental program attracted many foreign students, including many Americans, who sought to attain professional qualifications in the new discipline. Having mastered the elements of the new psychology and the structure of the German university system, they returned home to create their own laboratories and PhD programs in psychology.

Wundt's distinctive form of scientific psychology was eventually displaced within Germany as rival programs were created at other German universities. As the 20th century advanced, German psychology, which faced increasing opposition from the philosophical community, developed into precisely the type of applied discipline that Wundt feared it would become. In a sense Wundt's fate was like that of the sorcerer's apprentice. In creating a form of scientific psychology based upon laboratory science, he unleashed powerful forces that he was unable to control—forces that, over a few generations, radically transformed the discipline (Danziger, 1990, p. 34).

PSYCHOLOGY IN GERMANY BEFORE WUNDT

Psychology had been recognized as a distinctive field of inquiry long before the creation of Wundt's laboratory, and academic philosophers in various countries had offered courses in the subject. For example, psychology was offered as a course



Wilhelm Wundt (center) with colleagues and students in the Leipzig laboratory.

at Marischal College, Aberdeen, in 1755 under the title “pneumology” (Robinson, 1986). Christian Wolff (1679–1754), professor of mathematics at the University of Halle, popularized the use of the term *psychology* in Europe in the 18th century. He distinguished between rational psychology, concerned with rationally demonstrable principles about the human soul (such as its simplicity), and empirical psychology, concerned with the empirical description and measurement of psychological faculties such as sensation, memory, and intellect (Wolff, 1732, 1734). However, Kant rejected the notion that there could be rationally demonstrable knowledge of human psychology analogous to that of logic and mathematics and famously denied that empirical psychology could attain the status of a genuine science of quantified dynamical laws.

Johann Friedrich Herbart: Dynamic Psychology

Johann Friedrich Herbart (1776–1841), who succeeded Kant as professor of philosophy at the University of Königsberg in 1809, tried to prove him wrong. He developed an elaborate quantified dynamical theory of the “movement” of ideas in *A Textbook of Psychology* (1816) and *Psychology as a Science Based Upon Experience, Metaphysics, and Mathematics* (1824–1825).

Herbart advanced an associationist psychology based upon a postulated system of ideas possessed of attractive and repulsive forces, which strive to attain

dynamic equilibrium. He claimed that ideas are attracted to consciousness via effort and are repelled from consciousness when they conflict with ideas that constitute the current **apperceptive mass** of consciousness: the constellation of connected elementary mental representations that constitute the current object of apperception or focused attention.

Herbart claimed that his theory provided an account of the apparent spontaneity of thought, in the fashion that Newton's gravitational theory provided an account of the apparent "wandering" motion of the planets. In both cases, apparently irregular behavior was shown to be a determinate consequence of fixed mathematical laws. According to Herbart, ideas are never lost completely, but are merely repressed below a **threshold of consciousness**. The repressed ideas can sum their weaker energies to gain sufficient strength to force their way into consciousness, displacing the original apperceptive mass. In this fashion certain ideas "pop" into one's mind apparently unheralded, and certain thoughts keep recurring against one's will.

Herbart's psychological theory had all the trappings of a scientific theory. It was presented as a series of mathematical equations, such as the following equation governing the threshold of consciousness:

Among the many, and for the most part, very complicated laws underlying the movements of concepts, the following is the simplest:

While the arrested portion of the concept sinks, the sinking part is at every moment proportional to the part unsuppressed.

By this it is possible to calculate the whole course of the sinking even to the statical point.

Mathematically, the above law may be expressed: $\sigma = S(1 - e^{-t})$ in which S = the aggregate amount suppressed, t = the time elapsed during the encounter, σ = the suppressed portion of all the concepts in the time indicated by t .

—(1816/1891, p. 395)

Unfortunately, Herbart did not specify empirical measures for his central constructs, so his theory was virtually impossible to evaluate empirically. Yet it served as a rich source of theoretical psychological concepts, such as the repression of ideas and cognitive equilibrium, which were later exploited by Freud and the social psychologist Leon Festinger (1919–1989). Although he maintained that psychology could become a dynamical and mathematical science like Newtonian physics, Herbart denied the possibility of an experimental psychology (since he claimed that ideas could not be individually isolated from the dynamical systems in which they occur) and an introspective psychology (since he claimed that many ideas are unconscious).

Herbart's follower, Moritz Wilhelm Drobisch (1802–1896), added an important new element to his theory in *Experimental Psychology According to the Method of*

Natural Science (1842). Drobisch noted that the continued operation of perception and memory ensures that any system of ideas will remain in a state of disequilibrium. Organisms experience a state of disequilibrium as unpleasant and are motivated to regain a state of equilibrium. This notion of **homeostatic motivation**, according to which organisms are motivated to eliminate states of disequilibrium that are experienced as unpleasant or painful, played a central role in the later theories of Freud and the behaviorist psychologist Clark L. Hull (1884–1952).

Herbart applied his abstract mathematical theory of ideas to the field of education. He claimed that novel ideas introduced to students should be consistent with and related to the apperceptive mass of previously mastered ideas. He held that learning occurs when new representational elements are associated with the apperceptive mass through **assimilation** (in which case the new representational element is integrated with the apperceptive mass) or **accommodation** (in which case the apperceptive mass is adjusted to incorporate the new representational element). Herbart is often treated as the founder of educational psychology and anticipator of Piaget's theory of cognitive development (which was based upon the concepts of assimilation and accommodation).

Herbart's theory was the dominant force in psychology in Germany when Wundt developed his program of physiological psychology in the late 19th century (Titchener, 1925), and the University of Leipzig was the center of Herbartian psychology (with Drobisch as its head) when Wundt took up his position there as professor of philosophy in 1875.

WILHELM WUNDT: PHYSIOLOGICAL PSYCHOLOGY

Wilhelm Maximilian Wundt was born in the village of Neckarau in the German principality of Baden, the fourth child of a Lutheran minister, Maximilian Wundt, and his wife, Marie Frederike. He came from a distinguished family that included university presidents and professors, scientists, physicians, government administrators, and theologians. After a poor academic start (he hated school, failed his classes, and was thought by his teachers to be ill-suited for any demanding professional career), Wundt excelled as a medical student at the University of Heidelberg and received his medical degree (with honors) in 1855.

Despite his academic success, Wundt had no interest in pursuing a professional career in medicine. He attributed his lack of interest to personal doubts about his own competence (Wundt, 1920, p. 99) and recounted an anecdote about his early days as a medical intern, when he was so tired that he accidentally gave a patient iodine instead of a narcotic (narrated in Diamond, 1980, p. 21). His interests turned to physiology, and he studied with Müller and du Bois-Reymond at the University of Berlin in 1856. He received his second doctoral degree in physiology in 1857 and returned to the University of Heidelberg as a lecturer that

same year. Wundt's first course on experimental physiology was conducted in his mother's apartment. It attracted only four students, which was a financial as well as a professional disappointment, since in those days a lecturer's salary was funded from student fees. In 1858, he was appointed as an assistant to Helmholtz, the new head of the Institute of Physiology at the University of Heidelberg, where he served until 1864.

The assistantship turned out to be somewhat of a disappointment for Wundt. It required him to teach introductory courses on physiology and laboratory methods to medical students, and Wundt had little opportunity to work with Helmholtz on his research. However, he developed his first course on "psychology as a natural science" in 1862 (Bringmann et al., 1975) and published *Contributions Toward a Theory of Sense Perception* in 1862 and *Lectures on Human and Animal Psychology* in 1863. Wundt continued Helmholtz's measurement of neural transmission and calculated the time taken for the transmission of nerve impulses from the sense organs through the nervous system to the musculature. He identified a temporal remainder not accounted for by simple transmission, which he attributed to mental processes such as choice and volition (Blumenthal, 1985a). This type of theoretical inference about mental processes on the basis of reaction-time measurements became characteristic of Wundt's later experimental research in psychology.

Wundt resigned his position at the University of Heidelberg in 1864, supporting himself and his private psychological "institute" from his book royalties. He had created his own apparatus for measuring reaction time some years earlier and began to assemble a collection of laboratory instruments, such as chronoscopes, for measuring time intervals; kymographs, for making graphical records; and tachistoscopes, for the very brief presentation of visual stimuli. He returned to the University of Heidelberg from 1871 to 1874, using his private laboratory to support the courses in experimental physiology he was required to teach at the university (Bringmann, Bringmann, & Cottrell, 1976).

In 1873 and 1874 Wundt published the first edition of his two-volume *Principles of Physiological Psychology*. This work, which perhaps deserves to be classified as the first real textbook of experimental psychology, was revised and expanded in 1880, 1887, and 1893 and published as three-volume editions in 1902–1903 and 1908–1911. It constituted Wundt's self-conscious attempt to mark out physiological psychology as "a new domain of science," independent of but related to physiology and philosophy. Although about two thirds of the work was devoted to the physiology of the nervous system and sense organs, it was an instant international success and was favorably reviewed by leading academics such as William James.

After he failed to secure Helmholtz's vacated chair at the University of Heidelberg (when Helmholtz moved to the University of Berlin), Wundt took up the position of professor of inductive philosophy at the University of Zurich from 1874 to 1875. In 1875, he accepted a chair in philosophy at the University of

Leipzig, the largest university in Germany at the time, where he founded the Institute of Experimental Psychology in 1879. He remained there until his retirement in 1917.

Wundt supervised 186 dissertations at Leipzig between 1876 and 1917, of which 116 were psychological (the other 70 were philosophical or historical). He was by all accounts a popular teacher of a popular subject and was an indefatigable worker (Robinson, 1987). He published many works in both psychology and philosophy, including the 10-volume *Völkerpsychologie* (1900–1920). He died in 1920 shortly after publishing his autobiography (Wundt, 1920).

The Leipzig Laboratory

Wundt's new scientific discipline began modestly enough. The first course he taught at Leipzig was on physiological psychology. Some of the demonstrations and practicals relating to the course were conducted in a storeroom provided by the university in the summer of 1876. The room (promised to Wundt in 1875) was located in an unpretentious structure called the *Konvikt* building, which convicts had erected to serve as a cafeteria for poor students. As Wundt's courses in physiological psychology became more and more popular, his "psychological laboratory" came to occupy more and more rooms in the building.



Convict Building, Leipzig University (location of Wundt's first laboratory).

The year 1879 is conventionally designated as the one in which psychology was founded in Germany as an institutional scientific discipline, because it was in the winter semester of that year that students attending Wundt's Monday evening seminar began to develop their experimental projects in the Konvikt rooms; these later became the subjects of their PhD dissertations and academic publications in psychology:

From the Fall of 1879 on, individual students began to occupy themselves in this room in the refectory building with experimental projects.

—(Wundt, 1909, p. 118, cited in Bringmann, Voss, & Ungerer, 1997, p. 128)

One of these students was Max Friedrich (1856–1887), the first student in Wundt's new “practical seminar” and the first to be awarded a PhD in psychology at Leipzig (Tinker, 1932). Friedrich began his study “On the Duration of Apperception During Simple and Complex Ideas” in the winter of 1879–1880. Another was G. Stanley Hall (1844–1924), the first American student to visit Wundt's Leipzig laboratory. Both Hall and Wundt served as subjects in Friedrich's experiments (along with Friedrich himself).

Wundt managed to improve his situation when the University of Breslau made him a lucrative job offer in 1883. The Leipzig administration was anxious to retain him, and Wundt was able to set conditions for remaining. The university increased his salary and authorized funds for the expansion and improvement of the Konvikt building facilities. Wundt's “Institute for Experimental Psychology” was officially listed in the university catalogue and provided with a regular annual budget. The laboratory continued to expand and in 1893 moved to a well-equipped 11-room facility in a classroom building. In 1897, 18 years after its founding, the now famous Institute of Psychology moved to specially designed rooms on the top floor of a brand-new building. By this time Wundt's new science of experimental psychology and associated PhD program were well established, attracting students and visitors from all over the world. His program demonstrated that a systematic scientific psychology, sustained by a social collective of teachers and students engaged in a common research agenda, was indeed a “practical possibility” (Danziger, 1980, p. 106).

Wundt was most active as an experimentalist during his early years at Heidelberg. He took a lively and controlling interest in the work of the Leipzig laboratory in the 1870s and 1880s, but at the end of this period delegated the day-to-day direction of the laboratory to a variety of assistants, such as James McKeen Cattell (1860–1944), Oswald Külpe (1862–1915), and Wilhelm Wirth (1876–1952), who became co-director of the laboratory in 1904 (Schröder, 1997). Wundt remained a critical observer of and commentator on experimental work until the early 1890s, when he withdrew from laboratory work altogether. He became increasingly preoccupied with the development of the theoretical components of

his psychology, although from time to time he published polemical defenses of his own conception of psychological experimentation (Wundt, 1907) and scientific psychology (Wundt, 1913).

In 1881 Wundt established *Philosophical Studies* (*Philosophische Studien*), the first journal dedicated exclusively to psychological research and the first to regularly publish experimental studies in psychology. He edited *Philosophical Studies* until 1902, when he relinquished the editorship to Wilhelm Wirth. The journal was retitled *Psychological Studies* (*Psychologische Studien*) in 1906. In the early years the journal mainly reported the experimental output of the Leipzig laboratory; one of the first reports published was Friedrich's PhD dissertation study on apperception.

Physiological Psychology

Herbart was the first to develop a theoretical system of quantified dynamical laws in psychology, but denied that mental states and processes could be investigated experimentally. Fechner was the first to develop quantified psychophysical laws based upon rigorous experiments, but did not extend experimentation to the exploration of purely mental states and processes. Wundt was the first to apply the experimental methods of physiology to those mental states and processes (such as thought, emotion, and the will) that were formerly the exclusive domain of philosophers and to establish the institutional resources necessary to develop the academic discipline of scientific psychology, such as a funded laboratory, PhD program, textbook, and journal.

Wundt's experimental psychology developed out of the German tradition of experimental physiology, but he insisted that his experimental psychology was distinct from it. He affirmed the reality of "psychic causality" and the autonomy of psychological explanation. Wundt acknowledged that "the facts of consciousness always presuppose, as their physiological substrate, complex nerve processes" (1902–1903/1904, p. 321), but denied that psychological principles could be reductively explained in terms of physiological principles. He consequently rejected the idea that the goal of physiological psychology was to "derive or explain the phenomena of the mental from those of physical life" (1902–1903/1904, p. 2).

Experimental Methods

Wundt promoted an experimental psychology of **immediate experience**, in contrast to **mediate experience**, which he treated as the subject matter of natural science. As Wundt put it, natural science deals with theoretically interpreted experience of the "outer" world, whereas psychology deals with "the facts of immediate

experience in relation to the perceiving subject himself" (Mischel, 1970, p. 5). Wundt believed that he could make the study of conscious experience an exact science by rigorously controlling experimental conditions. His experimental program was based upon the assumption that rigorously controlled physical stimuli reliably generate the same sensational and perceptual responses in trained experimental observers, and he treated inter- and intrasubject replicability of results as the primary measure of the scientific objectivity of his experiments.

Wundt rejected the traditional philosophical conception of introspection as a form of "inner perception" (*innere Wahrnehmung*), which he condemned in the first issue of *Philosophical Studies*. He claimed that "there is . . . no such thing as an 'inner sense' which can be regarded as an organ of introspection" (1897/1902, p. 2). In contrast to "pure" or "armchair" self-observation, in which subjects simply describe or interpret their experience, he advocated the method of **experimental self-observation** (*experimentelle Selbstbeobachtung*), in which trained subjects provide concurrent commentaries on their conscious experience under rigorously controlled experimental conditions, which he hoped would avoid the distorting effects of intellectual reflection and reconstructive memory (Blumenthal, 1985a).

However, only a very small proportion of the work done in Wundt's laboratory involved the direct reporting of experience based upon experimental self-observation. Of the 180 experimental reports of studies conducted in Wundt's laboratory between 1883 and 1903 surveyed by Danziger (1979), only four contained introspective reports. Many of the experimental studies produced in Wundt's laboratory and published in *Philosophical Studies* were studies of sensation and perception, in which subjects made judgments about the quality or intensity of sensations, discriminated color differences and contrasts, estimated spatial positions and temporal intervals, or made determinations of simultaneity and succession (Danziger, 1990). Many were developments of Fechner's psychophysical studies, in which changes in physical stimuli were correlated with the perceived intensity of visual, tactile, and auditory sensations.

Other experiments employed measures of reaction time and were developments of the **complication experiments** pioneered by the Dutch physiologist Franciscus Cornelius Donders (1818–1889), in which the time taken for components of a complex task is calculated by subtraction of the measured time taken for other components of the task. Donders had used reaction times to simple and complex stimuli to measure the time taken to perform a variety of mental tasks. By subtracting simple reaction time for a response to a single stimulus from the time taken to discriminate a predesignated stimulus from a variety of presented stimuli, he estimated the time taken to perform the mental process of discrimination. By subtracting discrimination and simple reaction time from the time taken to choose a predesignated reaction to presented stimuli, he estimated choice reaction time (Donders, 1868). This experimental technique for measuring the duration of

postulated mental processes became known as **mental chronometry** and was a common feature of the Leipzig laboratory program until the turn of the century.

Wundt and his students created more complex stimuli, requiring subjects to respond to visual stimuli of a specific color, intensity, or duration, for example, and more complex responses, requiring subjects to make concurrent responses to visual and auditory stimuli, for example, by locating the position of a pendulum when a certain sound is heard. In this fashion, Wundt hoped to measure the time taken by mental mediating processes and to determine the nature of processes such as attention, judgment, memory, and inference (O'Donnell, 1985).

Wundt had conducted experiments of this sort since the early 1860s, using his own specially designed “thought meter”—a pendulum clock hooked up to bells and a calibrated scale (Wundt, 1862b). He used this instrument to determine the time it takes to shift attention from one “thought” or perceived stimulus, such as the ringing of a bell, to another, such as the position of a moving pointer (supposedly one tenth of a second), and the maximum number of “thoughts” or stimuli that could be attended to at one time (supposedly only one). The complication experiments conducted in Wundt’s laboratory went beyond the exploration of basic perceptual processes to the study of selective attention. James McKeen Cattell, one of Wundt’s American students, conducted a series of experiments on the identification of letters. He determined that reaction time for the naming of letters decreases as more letters are presented and that reading (aloud) times for connected letters and words are shorter than those for unconnected letters and words. His work was published in *Philosophical Studies* in 1885.

Apart from the occasional use of experimental self-observation, the types of studies conducted in the Leipzig laboratory did not differ radically from many of those in physiological laboratories. “Psychological” topics such as sensation and perception had formed part of the theoretical and experimental repertoire of physiologists since von Haller and Müller, and (despite his avowals to the contrary) to a significant degree Wundt simply appropriated that area of experimental physiology concerned with psychological dependent variables.

Wundt’s Psychology

Like most other German scientists, Wundt accepted the evolution of species, although his own position was closer to Lamarck’s and Spencer’s than to Darwin’s. He endorsed the inheritance of acquired characteristics and conceived of individual and species development as intrinsically teleological—as a goal-directed process of differentiation. He agreed with Morgan that all animal psychology and behavior “can be accounted for by the simple laws of association” (Wundt, 1863/1894, p. 350), but (also like Morgan) denied that all human psychology

and behavior could be explained in this fashion. Although he acknowledged that humans do passively associate ideas and behavior in accord with familiar principles of similarity, contiguity, and repetition, he maintained that the “elements” of human consciousness and cognition are not compounded in the aggregative fashion of associationist psychology, but are formed into integrated and unified configurations through the voluntary action of the will. He consequently denied that there is strong continuity between human cognitive processes and the associative processes common to humans and animals.

Wundt employed the term **apperception** to designate the creative and selective attentional processes that he believed are responsible for the configuration of conscious mental states. He claimed that apperception was the evolutionary advance in mental development that distinguished humans from animals and made possible the development of complex cultural forms of human mentality such as language, myth, and custom. Wundt held that the distinctive property of apperception is “creative synthesis” (*schöpferische Synthese*) and maintained that all other human psychological processes, such as perception, thought, and memory, are controlled by this central process, which he located in the frontal lobes of the brain (Blumenthal, 1975). According to Wundt, sensory elements are passively apprehended in the field of consciousness, but only some of these elements become the focus of attention in the selective configuration of mental states. Because of this emphasis on the voluntary, selective, and creative nature of the central control process of apperception, Wundt characterized his theoretical psychology as **voluntaristic psychology** (Wundt, 1896).

The two basic elements of Wundt’s theoretical system were sensations and feelings (with volition conceived of as a form of feeling); he held that they admitted of two fundamental properties, quality and quantity. Wundt analyzed feelings as varying along three dimensions: pleasant versus unpleasant, high versus low arousal, and concentrated versus relaxed attention. His Leipzig students devoted much time to the exploration of his **tri-dimensional theory of feeling**, but eventually abandoned it as unworkable, although later researchers (employing factor analysis) claimed to have identified affective dimensions similar to Wundt’s (Osgood, Suci, & Tannenbaum, 1957; Schosberg, 1954).

Wundt (1912/1973, p. 44) consistently maintained that

The whole task of psychology can be summed up in these two problems: (1) what are the elements of consciousness? (2) what combinations do these elements undergo, and what laws govern these combinations?

However, he did not conceive of this task in terms of the determination of laws of the association of independent conscious elements, although later critics complained about the “elementalism” of his program. Wundt always stressed that consciousness is a *process* composed of constituent processes and that the

“elements” of consciousness are intrinsic components of complex configurations, which can be identified or inferred only via experimental analysis and abstraction.

According to Wundt’s **principle of psychical resultants** (also known as the *principle of creative synthesis*), the attributes of psychological configurations that are the product of apperception are distinct from the mere aggregation of the attributes of the elements from which they are configured. This principle was held to apply to quantities (e.g., the perceived intensity of a sensory stimulus) as well as qualities (e.g., the perception of spatial relations). According to Wundt, psychological configurations, such as the perception of a musical chord or understanding of a sentence, have emergent properties that cannot be reduced to the mere aggregation of elemental properties:

Every psychological compound shows attributes which may indeed be understood from the attributes of its elements after these elements have once been presented, but which are by no means to be looked upon as the mere sum of the attributes of these elements.

—(1897/1902, p. 321)

Wundt originally followed Mill and Helmholtz in accounting for the emergent properties of psychological configurations in terms of unconscious cognitive inference, but later came to treat them as a product of the creative “fusion” of the elements of consciousness. He also claimed that this creative fusion accounts for the integration of motor movements in goal-directed behavior.

Wundt did not merely claim with Mill and Helmholtz that complex ideas have emergent properties that cannot be reduced to the properties of their sensational components, in the fashion that the properties of molecular compounds such as water cannot be reduced to the properties of their atomic components, such as hydrogen and oxygen. He rejected Mill’s account of the formation of complex ideas as a form of mental chemistry, because he claimed that Mill had neglected the “special creative character of psychic synthesis” (1902, cited in Blumenthal, 1975). Wundt maintained that the “elements” of psychological configurations cannot be identified and isolated independently of their configuration, unlike atomic elements such as hydrogen and oxygen, which can be identified and isolated independently of molecular compounds such as water.

Wundt held that the “elements” of consciousness are relational rather than atomistic. According to Wundt’s **principle of psychical relations**, the nature and identity of the elements of psychological configurations are determined by their relational location within psychological configurations:

Every single psychical content receives its significance from the relations in which it stands to other psychical contents.

—(1897/1902, p. 323)

In the production of psychological configurations, the significance of the elements attended to derives from their apperceived relation to other elements in the psychological configuration. For example, Wundt claimed that words do not have meaning in isolation, but only via their role in configured sentences (1900, p. 37). This relational conception of the “elements” of perception and cognition later became the foundational principle of the Berlin school of Gestalt psychology represented by Max Wertheimer (1880–1943), Wolfgang Köhler (1887–1967), and Kurt Koffka (1886–1941).

Many of Wundt’s experimental studies of apperception were precursors of contemporary research in cognitive psychology on attention span and short-term memory (Blumenthal, 1985a; Leahey, 1979). Wundt had originally thought it possible to attend to only one thought or stimulus at a time, but experimental studies in the Leipzig laboratory established that about six or seven items could be simultaneously attended to. These experimental studies, which were developed by Wilhelm Wirth in *The Experimental Analysis of the Phenomena of Consciousness* (1908), anticipated George Miller’s (1956) classic study of the restriction of short-term memory capacity to about seven units. Wundt’s studies also anticipated Miller’s finding that the “chunking” of these “elements” into larger meaningful units can increase the capacity of short-term memory (by forming letters into words or numbers into ordered sequences, for example).

Wundt claimed that apperception plays a critical role in the perception of space and time, the operation of imagination and reasoning, and linguistic processing. He explained linguistic performance in terms of the transformation of thought configurations into symbolic representations in language. According to Wundt, a speaker apperceives a configured idea and selects a sequence of linguistic symbols to express it; a listener analyzes the speaker’s linguistic production in an attempt to apperceive the original configured idea of the speaker. The process of communication can go astray via the failure of the speaker to properly express the original configured idea, or of the listener to reconstruct it. Wundt noted that the original configured idea can be expressed by the speaker (or reconstructed by the listener) in a variety of different linguistic forms, which he thought explained how we can often remember the meaning of a verbal communication (or of a piece of prose or poetry) after we have forgotten the specific sentences used to express its meaning (Wundt, 1900). Wundt’s account of linguistic processing anticipated later developments in psycholinguistics and the transformational grammar of Noam Chomsky (1928–); he invented the tree diagrams representing sentence structure later employed by many linguists, including Chomsky (1957).

Wundt also suggested that disruption of the attentional mechanisms of apperception might be the source of some psychological disorders—a suggestion developed by his student and friend Emil Kraepelin (1856–1926), one of the pioneers of German scientific psychiatry.

Völkerpsychologie

By the early 1900s Wundt had lost interest in laboratory work, although he never abandoned his commitment to experimental psychology and defended it vigorously against philosophical critics in his 1913 book *Psychology Struggling for Survival*. He stopped serving as editor of *Philosophical Studies* in 1902 and turned over the everyday operations of the Institute of Psychology to Wilhelm Wirth in 1908. He devoted most of his remaining years to his 10-volume *Völkerpsychologie* (1900–1920), variously translated as “social psychology,” “cultural psychology,” or “folk psychology,” a comparative-historical study of the “mental products” of social communities, such as language, myth, and custom.

The idea of a psychology grounded in social community had been suggested by Herbart and developed by Humboldt. Moritz Lazarus (1824–1903) first articulated the idea of a special discipline devoted to the comparative and historical study of the mental products of social communities in a paper titled “On the Concept and Possibility of a *Völkerpsychologie*” in 1851 (Jahoda, 1997). Lazarus founded the journal *Zeitschrift für Völkerpsychologie und Sprachwissenschaft* with Hajm Steinthal (1823–1899) in 1860. However, the notion of a comparative and developmental psychology grounded in cultural and historical differences in social community has a much longer history. Wundt’s own project represented the continuation of a tradition that can be traced back to Vico and Herder and that found partial expression in Kant’s anthropology and Mill’s ethology.

Many historians of psychology have claimed that Wundt denied the possibility of studying “higher” psychological states and processes experimentally and have asserted that Wundt believed these could be studied only by the naturalistic comparative-historical methods of *Völkerpsychologie* (Farr, 1996; Shook, 1995). Yet this is a misrepresentation of Wundt’s position. He did lose interest in laboratory work at Leipzig, but did not turn to the naturalistic observational methods of *Völkerpsychologie* in reaction to the problems of laboratory experimentation directed toward “higher” psychological states and processes. Wundt became interested in *Völkerpsychologie* very early in his career. He offered his first course on the subject (entitled *Anthropologie*) in 1859, during his second year of teaching at Heidelberg (Leary, 1979) and detailed the project of *Völkerpsychologie* in his *Lectures on Animal and Human Psychology* in 1863. He claimed that naturalistic comparative-historical observation, the method of Darwin, is the best method for studying processes of development, whether biological, psychological, or social.

Wundt supervised experimental projects on “higher” psychological processes such as thought and memory in the Leipzig laboratory—for example, Harry K. Wolfe’s “Studies on the Memory of Tones” and Edward W. Scripture’s “Thinking and Feeling” (Benjamin et al., 1992). Apperception, which Wundt held to be the central control process governing all other human psychological processes (Blumenthal, 1975), was a major focus of the *Völkerpsychologie*, but

was also the subject of experimental research in the Leipzig laboratory: The very first experimental study conducted in Wundt's laboratory was Friedrich's study of apperception.

Wundt saw the naturalistic observational methods of *Völkerpsychologie* as a *complement* to those of experimental psychology:

Psychological analysis of the most general mental products, such as language, mythological ideas, and laws of custom, is to be regarded as an aid to the understanding of all the more complicated psychical processes.

—(1897/1902, p. 10)

According to Wundt, experimental psychology can never be a science of pure observation, because unlike physical sciences that deal with “relatively permanent objects of nature” that are independent of human consciousness, it is restricted to the study of fleeting psychological “processes” that are dependent upon human consciousness. This was the reason Wundt insisted upon rigorous control in the experimental investigation of conscious psychological processes. However, he claimed that the mental products of social communities that form the subject matter of *Völkerpsychologie* are sufficiently akin to the “relatively permanent objects of nature” to admit of something analogous to pure observation, “inasmuch as they possess . . . attributes of relative permanence, and independence of the observer” (1897/1902, p. 22).

Wundt maintained that social-psychological facts about language, myth, and custom could serve as observational grounds for inferences about psychological processes:

The origin and development of these products depend in every case on general psychological conditions which may be inferred from their objective attributes. Psychological analysis can, consequently, explain the psychical processes operative in their formation and development.

—(1897/1902, p. 23)

He claimed that differences in psychological processes could be inferred from differences in the linguistic, mythical, and cultural products of social communities. For example, he suggested that differences in psychological motives could be inferred from the different types of word orderings of sentences in different languages (1912/1970, p. 28).

Wundt held that psychological processes and products could be investigated by both laboratory experimentation *and* the naturalistic comparative-historical methods of *Völkerpsychologie*. When he claimed that the subject matter of *Völkerpsychologie* is “unapproachable by means of experiment in the common acceptance of the term” (1897, p. 23), he meant that the historical development of social

forms of language, myth, and custom cannot be investigated via experimental self-observation, since

individual consciousness is wholly incapable of giving us a history of the development of human thought, for it is conditioned by an earlier history concerning which it cannot of itself give us any knowledge.

—(1916, p. 3)

Wundt's attitude toward experimentation in relation to *Volkerpsychologie* was analogous to his attitude toward experimentation in child and animal psychology. He never denied that experimental methods of intervention and manipulation could be employed in these fields of psychology, but claimed that experimental *introspection* was of limited utility:

Results of experiment are here matters of objective observation only, and the experimental method accordingly loses the peculiar significance which it possesses as an instrument of introspection.

—(1902–1903, p. 5)

Wundt's 10 volumes of *Völkerpsychologie* constituted a rather disappointing compilation of largely anecdotal ethnographic accounts of myths, rituals, religions, and customs, which provided doubtful evidential support for his theoretical speculations about social and cognitive development (Jahoda, 1997). The most interesting are the first two volumes devoted to language, which were also the most successful. Wundt rejected the Herbartian conception of language as a set of linguistic elements compounded according to principles of association (Paul, 1880) and maintained that language is creatively generated in accord with abstract rules governing the production of sentences (Blumenthal, 1970; Mischel, 1970). Despite its limitations, Wundt's *Völkerpsychologie* indicated the potential of comparative historical studies of the social foundations of thought and language, of the type later developed by Lev Vygotsky (1896–1934) and his colleagues in Russia (Cole, 1996).

Wundt's Legacy

Wundt had many students but few intellectual disciples and is not generally remembered for his substantive contributions to theoretical or experimental psychology—there are no enduring psychological laws or principles named after him. His main achievement was the establishment of a research community of psychologists working on a common set of experimental problems, who disseminated their results through PhD dissertations and journal publications (Danziger, 1990).

Wundt established the viability of scientific psychology and trained a generation of PhD certified professional psychologists who set about reproducing their own research communities as they instituted laboratories in Europe and America, although their own research programs and practical agendas often differed radically from Wundt's own vision of scientific psychology.

Wundt established psychology as an academic discipline in Germany in the face of vigorous opposition from the philosophers of his day, who claimed that too much self-observation would drive young people to insanity. Although he insisted that scientific psychology was distinct from physiology, it was never his intention to establish psychology as an academic discipline independent of philosophy. Wundt's chair in Leipzig was in philosophy, and he carried out his teaching and supervisory duties in philosophy as enthusiastically as he did those in psychology, publishing works on logic, ethics, metaphysics, and philosophy of science along with his psychological output. He considered scientific psychology to be intimately related to philosophy and hoped that it would transform and reinvigorate late-19th-century German philosophy. He considered psychology to be "both a part of the science of philosophy and an empirical *Geisteswissenschaft*": "its value for both philosophy and the empirical special sciences resides in its being the main negotiator between them" (1913, p. 32).

This commitment did not save him from the criticisms of philosophers like Wilhelm Windelband (1848–1915), who complained that

For a time it was thought in Germany that one was close to being qualified for a philosophical chair as soon as one had learned to press electric buttons in a methodological way, and as soon as one could numerically prove by means of well-ordered and tabulated series of experiments that some people get ideas more quickly than others.

—(quoted in Kusch, 1995, p. 171)

Wundt made a heroic (if unsuccessful) effort to integrate the principles of experimental psychology and *Völkerpsychologie* and conceived of psychology as a **propaedeutic science** (Blumenthal, 1985a) that provided the foundation of both the natural and social or human sciences. However, later psychologists followed Dilthey in maintaining that the natural and social or human sciences are fundamentally different, and they opted for the natural scientific version of scientific psychology, with its promise of technological application.

Wundt's lasting achievement was the establishment of psychology as an autonomous scientific discipline. He trained generations of German and foreign students in the new science; they went on to found their own laboratories and psychology programs in Germany, the Americas, India, Russia, and Asia. Wundt's Indian students produced a large commemorative volume in 1932, the centenary of Wundt's birth, and his Russian and Japanese students constructed replicas of Wundt's Leipzig laboratory in Moscow in 1912 and in Tokyo in 1920 (Blumenthal, 1975).

After Wundt's death in 1920, some of his former students, such as Felix Krueger (1874–1948), Friedrich Sander (1880–1971), and Wilhelm Wirth continued to develop his configurational psychology at Leipzig and reinstituted *Psychological Studies* (which ceased publication after Wundt's retirement in 1919) as *New Psychological Studies* (Blumenthal, 1975). They later became known as the Leipzig school of Gestalt psychology, although the more famous Berlin school of Gestalt psychology represented by Wertheimer, Köhler, and Koffka eclipsed their work.

Along the way Wundt's substantive achievements in cognitive psychology and his project for a comparative historical *Völkerpsychologie* were forgotten. This was because they were neglected by the Americans who studied under Wundt and who returned home to shape the development of the form of scientific psychology that eventually came to establish a virtual global hegemony. As American psychology expanded dramatically in the early 20th century, German psychology was emaciated by the economic depression of the 1920s. Leipzig University could not afford to purchase Wundt's last works for the university library, and a Japanese consortium purchased his personal library (Miyakawa, 1981). Many of Wundt's students were removed from their positions in Germany, Italy, and Russia by hostile Fascist and Marxist regimes (Blumenthal, 1985b).

Wundt's American Students

Most of Wundt's American students came to the Leipzig laboratory in search of academic qualifications that they could not get back home (only a few Ivy League institutions offered the PhD degree in the late 19th century, and fewer still in psychology) and the prestige and earning potential of a German degree. They returned with only the experimental skeleton and institutional structure of the new psychology. Two exceptions were G. Stanley Hall and Charles Judd (1873–1946). Hall persuaded Franz Boas (1858–1942) to visit Wundt in Leipzig and to offer a course on *Völkerpsychologie* when he returned to Clark University. Boas moved to Columbia University, where he founded the school of "cultural anthropology" that came to include such distinguished figures as Margaret Mead (1901–1978) and Ruth Benedict (1887–1948). Judd made a valiant attempt to promote an "institutional" form of social psychology based upon Wundt's *Völkerpsychologie* (Judd, 1926), but with little success. Eventually he abandoned the project and turned to educational psychology.

For those Americans who did not study with Wundt, their exposure to Wundt's psychology was generally filtered through the translations and expositions of his work produced by Edward B. Titchener (1867–1927), who interpreted his psychology in terms of the atomistic and associationist psychology of British empiricism, to which it bore little resemblance (Blumenthal, 1975; Leahey, 1981). Wundt's spirited defense of Germany in the First World War did little to promote

his theoretical system, but by that time the distinctive features of his psychology had already been lost to most Americans.

Still, Wundt's influence on the development of scientific psychology in America should not be underestimated. Thirty-three Americans completed their doctoral degrees under his supervision. Ten years after the founding of Wundt's laboratory in Leipzig, there were over 40 American laboratories, about a dozen of which were founded by his students (Benjamin et al., 1992). Wundt's American students included Frank Angell (1857–1939), who founded laboratory programs at Cornell University and Stanford University; James McKeen Cattell, who founded laboratory programs at the University of Pennsylvania and Columbia University; Walter Dill Scott (1869–1955), who founded the laboratory program at Northwestern University and was a pioneer of industrial psychology; and Lightner Witmer (1867–1956), who took over the laboratory at Pennsylvania (founded by Cattell) and instituted the first psychological clinic. Other laboratory programs were founded by Harry Kirke Wolfe (1858–1918) at the University of Nebraska; Edward Wheeler Scripture (1864–1945) at Yale University; Edward Aloysius Pace (1861–1938) at the Catholic University of America; and George Stratton (1865–1957) at the University of California at Berkeley.

GERMAN PSYCHOLOGY BEYOND LEIPZIG

Wundt's Leipzig program remained a dominant force in German psychology for many years, but was not the only laboratory-based psychology program developed in Germany at the end of the 19th and early 20th centuries. A number of other German universities created institutes of psychology, whose programs came to rival and eventually supersede Wundt's own version of physiological psychology.

Hermann Ebbinghaus: On Memory

Hermann Ebbinghaus (1850–1909) originally trained in philosophy at the University of Bonn. He traveled throughout Europe for a number of years, working as a part-time teacher and private tutor. He is reputed to have developed an interest in experimental psychology after reading a copy of Fechner's *Elements of Psychophysics* (1860), which he bought in a secondhand bookstore in Paris. The story may be apocryphal, but Ebbinghaus committed himself to the task of studying memory experimentally as Fechner had earlier committed himself to the task of studying sensation experimentally. He returned to Germany in 1880, where he served as an untenured instructor at the University of Berlin. He began a series of experimental studies on memory, defined as "learning, retention, association and reproduction" (1885, p. v), with himself as the single experimental subject.

To avoid the contamination of learning and memory by previous associations and meaningful relations, Ebbinghaus created lists of meaningless syllables (based upon consonant-vowel-consonant combinations) and used randomly selected combinations of these syllables as his stimulus material. On each learning trial, he looked at each syllable on a list for a fraction of a second, as measured by a metronome. He repeated this process every 15 seconds until he attained mastery of the list—when he was able to recall each syllable on a list without error. Ebbinghaus established that the number of repetitions required to learn a list increases with the length of a list: A list with 7 syllables requires only 1 repetition, a list with 16 requires 30 repetitions, and a list with 36 requires 55 repetitions. He demonstrated that memory deteriorates rapidly in the first few hours and days after learning, and much more slowly thereafter: over 50 percent of the material is forgotten in the 1st hour, and over 60 percent in the 1st day, with around 20 percent retained between the 2nd and the 30th day. Ebbinghaus also established that the time taken to relearn lists after initial exposure decreases as the number of original repetitions increases, demonstrating the importance of overlearning.

Twelve years after graduating from the University of Bonn, Ebbinghaus published *On Memory* (1885), which he dedicated to Fechner. This book described an elegant set of studies directed to specific hypotheses, in the scientific tradition of Gilbert's *On Magnetism* (1600) and Newton's *Opticks* (1704), and became an instant classic of experimental reportage. William James, who had little good to say about the work of most of the pioneers of German psychology, called Ebbinghaus one of Germany's "best men." As a result of the success of his book, Ebbinghaus was appointed professor extraordinarius at the University of Berlin.

On Memory stood the test of time better than most of Wundt's publications and initiated a long tradition of research on memory (Postman, 1968). However, Ebbinghaus did not continue his memory research at Berlin. His own interests turned to sensory and perceptual psychology, possibly because of the influence of Helmholtz, who held a chair in physics at the university at the time. Ebbinghaus founded an Institute of Experimental Psychology at the University of Berlin, but played only a minor role in the institutional development of German psychology. He published little else of note and devoted much of his energy to the editorship of the *Journal of Psychology and the Physiology of the Sense-Organs*, which he founded (with Arthur König) in 1890 as an alternative to Wundt's *Philosophical Studies*.

Possibly as a consequence of his lack of research output, he was not promoted to the chair in philosophy at the University of Berlin, which was given to Carl Stumpf (1848–1936) in 1894. Ebbinghaus spent the later years of his career at the universities of Breslau and Halle, where he published two extremely popular textbooks of psychology, *Principles of Psychology* (1897) and *An Elementary Textbook of Psychology* (1902), which ran into several editions. He also pioneered the use of completion tests for assessing the intelligence of children after a Breslau education committee asked him to advise them on the most effective way of structuring the

school day with a view to promoting productive learning. Alfred Binet (1857–1911) employed some of these tests in his intelligence scales, and they were later incorporated in the Stanford-Binet intelligence test.

Georg Elias Müller: The Experimentalist

Georg Elias Müller (1850–1934) was a native of Saxony who originally studied philosophy and history at the University of Leipzig. After service in the infantry during the Franco-Prussian war (1870–1871), he returned to Leipzig to study with Moritz Drobisch. Throughout his academic career, Müller followed Drobisch in defending Herbart's form of associationist psychology. On Drobisch's recommendation, he went to study at the University of Göttingen with Rudolf Hermann Lotze (1817–1881), who held the chair in philosophy formerly held by Herbart. After producing a thesis on attention under Lotze's supervision in 1873, Müller returned to the University of Leipzig, where he met Fechner and developed an interest in psychophysics.

In 1878 he published *Fundamentals of Psychophysics*, a critical evaluation of Fechner's *Elements of Psychophysics* (1860/1866). Fechner's own detailed response to his critique, *Revision of the Main Points of Psychophysics* (1882), established Müller's reputation, and shortly afterward he was appointed to the chair of philosophy at the University of Göttingen (when Lotze moved to the University of Berlin), where he remained until his retirement in 1921. Müller immediately set about founding a laboratory and PhD program in psychology, although he did not secure dedicated laboratory space until 1887 and (modest) funding until 1891 (Blumenthal, 1985b).

Müller was a dedicated experimentalist, whose lifelong commitment to methodological rigor, quantification, and instrumentation was possibly more representative of the spirit of modern scientific psychology than Wundt's own version (Müller had no interest in philosophy or *Völkerpsychologie*), which led William James to describe his contribution as "brutal." Müller's form of physiological psychology was also far more physiological than Wundt's. He was a reductive explanatory materialist, who insisted that scientific psychological theories must be grounded in physiological theories; his own psychological theories were based upon physiological theories of cortical blood supply and neural excitation (Blumenthal, 1985b).

Although he was not an innovative theorist, Müller was an enormously productive scholar, who developed the work of earlier theorists such as Ebbinghaus, Fechner, and Helmholtz (Behrens, 1978). He extended Fechner's program of psychophysics, making Göttingen a major center for psychophysical research. He introduced a number of methodological and statistical innovations and produced masterful studies of the psychophysics of lifted weights (Müller & Martin, 1889; Müller & Schumann, 1889).

Müller also developed the program of memory research that Ebbinghaus had initiated and established the research tradition of verbal learning (Blumenthal, 1985b). He and his student Friedrich Schumann (1863–1940), a PhD in physics who became Müller's first laboratory assistant in 1881, invented the memory drum (Müller & Schumann, 1893), which was for many years the standard instrument for the study of verbal learning and memory (Behrens, 1997). He also developed an early interference theory of memory with another of his students, Alfons Pilzecker (Müller & Pilzecker, 1900). Müller's extensive experimental studies of recall and recognition were summarized in his three-volume *Analysis of the Processes of Memory and Mental Representation* (1911–1913). He also developed and modified Ewald Hering's (1834–1918) "opponent-process" theory of visual perception and Wundt's theory of spatial localization.

Müller recognized the critical role of configurational properties (*Gestaltsqualitäten*) in the organization of perception, thought, and memory, but rejected Wundt's theory of creative synthesis. He based his own structural theory (*Komplextheorie*) upon traditional Herbartian principles of association (Behrens, 1997) and treated configurational properties as complex by-products of association (Blumenthal, 1985b). He was highly critical of the form of Gestalt psychology developed by Wertheimer, Köhler, and Koffka, which he dismissed as unoriginal and methodologically unsound in *Structure Theory and Gestalt Psychology* (1923). However, a number of Müller's students went on to develop their own versions of Gestalt psychology, such as Erich R. Jaensch (1883–1940), David Katz (1884–1953), Albert Michotte (1881–1965), and Edgar Rubin (1886–1951).

Müller played almost as influential a role as Wundt in the early development of experimental psychology in Germany and in establishing the 20th-century research traditions that grew out of it. However, he is rarely remembered in introductory texts and histories of psychology, although his work was extensively cited in Titchener's 1905 *Experimental Psychology*, second only to Wundt and Fechner (Behrens, 1997), and in Robert S. Woodworth's 1938 *Experimental Psychology*, second only to Wundt and more frequently than Ebbinghaus, Köhler, Pavlov, Thorndike, and Titchener (Blumenthal, 1985b). Few of Müller's works were published in or translated into English. He played a significant role in the creation of the German Society of Experimental Psychology in 1904 but was uninterested in developments in psychology outside Germany and contributed little to them.

Müller had less influence than Wundt on the institutional development of American psychology and had fewer American students. However, he had some very interesting ones, including a number of American women. Christine Ladd-Franklin (1847–1930), one of the first graduates of Vassar College, worked briefly with Müller on color perception in 1891. She also studied with Helmholtz and developed her own version of the "opponent-process" theory of color vision. Before she visited Müller in Germany, she fulfilled all the requirements for a graduate degree (in mathematics) at Johns Hopkins, but the university would not grant

her one because in its early years it denied degrees to women. She never managed to attain a full-time academic position (her marriage to Fabian Franklin, a Hopkins mathematician, precluded her from consideration for the few academic positions that were open to women in her day). Hopkins finally awarded her a degree in 1926, 44 years after she completed her studies and 4 years before she died (Scarborough & Furumoto, 1987). In later years she became a militant feminist.

Lillien Martin (1851–1943), another graduate of Vassar College, was a student of Müller's from 1894 to 1898. They published a joint study on the psychophysics of lifted weights that became a classic (Müller & Martin, 1899). She completed all the courses required for a degree, but never received one because the University of Göttingen prohibited women from graduating. When Martin returned to the United States, she got a job in the department of psychology at Stanford University. She continued to work on perception (and aesthetics) and became chair of the department in 1915 (Scarborough & Furumoto, 1987). After her retirement in 1916, she continued to travel and lecture on psychological topics, including (appropriately enough) aging. She died at the age of 91, after a brief dizzy spell.

Eleanor Gamble (1868–1933), who completed her PhD with Edward B. Titchener at Cornell on the psychophysics of smell, received a postdoctoral research grant to study with Müller in 1906. On her return, she became director of the Wellesley psychological laboratory, where she supervised psychological research for the next 25 years. Gilbert Haven Jones (1883–1966), an African American from Fort Mott, South Carolina, earned a doctoral degree in philosophy at the University of Jena in 1901, and studied with Müller for a few years before returning to the United States. He was appointed professor of philosophy and education at St. Augustine College in North Carolina, where he became the first African American with a doctoral degree to teach psychology. William McDougall (1871–1938), the “purposive” behaviorist and early pioneer of American social psychology, also studied with Müller, and Oswald Külpe (1862–1915), the founder of the Würzburg school of psychology, received his first degree at Göttingen.

Franz Brentano: Intentionality

Franz Brentano (1838–1917), the son of an Italian merchant immigrant to Germany, began training for the priesthood at the age of 17. He studied philosophy at the universities of Berlin and Munich, where he developed a lifelong interest in the philosophy of Aristotle. Brentano received his PhD from the University of Tübingen in 1862 for a dissertation on Aristotle and was ordained as a priest shortly afterward. He obtained a position at the University of Würzburg, where he proved to be a popular teacher of philosophy. Finding himself on the wrong side of the debate about the pope's infallibility, Brentano resigned from the priesthood and his academic position in 1873 (he felt obligated to since he had originally been appointed as a priest).

During the next few years Brentano worked on what became his magnum opus, *Psychology From an Empirical Standpoint*, which was published in 1874 (the same year as the second volume of Wundt's *Principles of Physiological Psychology*). He secured an academic appointment as professor of philosophy at the University of Vienna, where he remained for the next 20 years, retiring from the university in 1894. He was forced to resign his official position at the university in 1880 when he renounced the Catholic Church and married. He taught as an unpaid instructor (albeit a very popular one) in his later years (Baumgartner & Baumgartner, 1991).

Following Aristotle and Aquinas, Brentano treated *intentionality* as the distinctive “mark of the mental.” According to Brentano, mental states such as thoughts, memories, and emotions are intentionally directed upon objects of thought, memory, and emotion. For example, my thought that the Empire State Building is the tallest building in New York is *about* the Empire State Building, and my anger at Sarah for having borrowed my laptop without asking is directed *at* Sarah. As Brentano put it,

Every mental phenomenon is characterized by what the Scholastics of the Middle Ages called the intentional. . . . Every mental phenomenon includes something as object within itself, although they do not all do so in the same way. In presentation something is presented, in judgment something is affirmed or denied, in love loved, in hate hated, in desire desired, and so on.

—(1874/1995, p. 88)

Because of his focus on the intentionality of acts of perception, judgment, and desire, Brentano is often characterized as a proponent of **act psychology**, in contrast to Wundt's supposed **psychology of contents**, concerned with the elemental contents of consciousness. However, Wundt placed at least as much emphasis on mental acts as he did on mental contents and always insisted that experimental psychology was directed to the exploration of mental *processes*.

Like Wundt, Brentano was firmly committed to the utility of experimental methods in physiological psychology, which he employed in his own empirical research on color vision. He petitioned the University of Vienna for funding for a psychological laboratory, but with less success than Wundt at Leipzig. However, Brentano was less enthusiastic than Wundt about the potential of physiological psychology. Like Müller, he was an explanatory reductive materialist, who believed that causal explanatory understanding of psychological processes was dependent upon causal explanatory understanding of the physiological processes that ground them

Psychology . . . will never fulfill its task without the inclusion of physio-chemical processes and the identification of anatomic structures.

—(Brentano, 1982/1995, p. 3)

Since he believed that physiology had few established causal explanatory principles at its disposal in the late 19th century, and fewer still of relevance to the causal explanation of the “succession of psychic phenomena,” he thought the immediate prospects for an explanatory scientific psychology were poor. His classification of mental acts and their properties in *Psychology From an Empirical Standpoint* (1874/1995) was entirely descriptive.

Although he did not develop a substantive research program in experimental psychology, Brentano had some important and influential students. At the University of Würzburg they included Carl Stumpf (1848–1936), who was preferred over Ebbinghaus and Müller for the chair in philosophy at the University of Berlin. At the University of Vienna, they included Edmund Husserl (1859–1938), the founder of phenomenological philosophy and psychology; Christian von Ehrenfels (1859–1932), one of the founders of Gestalt psychology; and Sigmund Freud, the founder of psychoanalysis.

Carl Stumpf: The Berlin Institute of Experimental Psychology

Carl Stumpf discovered philosophy as a student of Brentano's at the University of Würzburg and completed his PhD at the University of Göttingen under Lotze's supervision in 1868. For a few years he studied for the ministry, but abandoned this projected vocation in 1870 over the issue of papal infallibility. He accepted a position as instructor in philosophy at the University of Göttingen (on Lotze's recommendation), where he worked with Weber and Fechner. He took over Brentano's position at the University of Würzburg when Brentano resigned in 1873. That same year Stumpf published a book on the psychology of visual perception that anticipated many of the themes of Gestalt psychology. In 1883 he published the first volume of his major work *The Psychology of Tone* (the second was published in 1890). An accomplished musician, Stumpf maintained a lifelong interest in the theory, practice, and psychology of music.

After holding positions at the universities of Prague, Halle, and Munich, Stumpf was appointed to the chair of philosophy at the University of Berlin in 1894, with an adjunct appointment as director of the Institute of Experimental Psychology (founded by Ebbinghaus). He did little empirical research and was skeptical of the potential of experimentation in psychology. Stumpf claimed that his own very limited training in experimental techniques was based upon a single course in chemistry, during which he narrowly avoided burning down the chemistry building (Stumpf, 1930), and allegedly boasted that he could carry all the laboratory apparatus he needed in a cigar box under his arm (O'Donnell, 1985).

However, Stumpf was an excellent organizer and administrator who quickly expanded the Institute of Experimental Psychology, focusing initially on space



Clever Hans.

perception and audition. He hired Friedrich Schumann, Müller's former research assistant, who began a 10-year program of research on time perception and the emergence of configurational properties in spatial perception and word recognition (Blumenthal, 1985b). As director, Stumpf instituted schools of medical, musical, military, and child psychology, as well as a center for the study of traditional music. Eventually the Berlin Institute of Experimental Psychology, which occupied the top floor of the former Imperial Palace in Berlin, came to rival Wundt's Leipzig Institute as the primary center for psychology in Germany.

Stumpf also promoted the development of industrial and other forms of applied psychology. Although he took little active part in applied research himself, he collaborated with one of his students Oskar Pfungst (1874–1933) in the investigation of the case of Clever Hans, a horse owned by Herr von Osten (1838–1909). The horse could apparently solve mathematical puzzles by tapping out answers when questioned by von Osten, a miraculous display that attracted daily crowds to von Osten's home in the northern suburbs of Berlin. Pfungst demonstrated that Clever Hans was incapable of performing his mathematical "solutions" when von Osten was not present and that the horse's "achievements" were responses to unconscious behavioral cues supplied by von Osten.

In 1896 Stumpf presided over the Third International Congress in Psychology, held at the University of Munich. In recognition of his achievements, and with the enthusiastic support of his friend William James, Stumpf was elected a fellow of the American Academy of Sciences and the American Psychological Association.

Stumpf was deeply critical of Wundt's psychology, which was roundly condemned at the Berlin Institute as too passive and elemental, albeit unjustly, given Wundt's emphasis on the active and constructive role of apperception in the creative synthesis of psychological configurations. Stumpf's hostility toward Wundt was a product of personal animosity and institutional rivalry, engendered by an acrimonious public dispute over the discrimination of tonal distances that reflected their radically different approach to empirical validation. Wundt's position was based upon controlled experimental studies employing naïve subjects, who were trained in introspective techniques but were not professional musicians, whereas Stumpf's position was based upon his own introspective experience, which he considered to be more sophisticated and consequently superior to that of musically untrained experimental subjects. Characteristically, James sided with his friend Stumpf against Wundt (Blumenthal, 1985b).

Although his own research had little direct influence, many of Stumpf's students had a major impact on subsequent developments in philosophy and psychology. Husserl was a student of Stumpf's from 1884 to 1886. The Gestalt psychologists Wertheimer, Köhler, and Koffka were all students of Stumpf's, as was Kurt Lewin (1890–1947), who later applied the principles of Gestalt psychology to social psychology (Lewin, 1948), and Max Meyer (1873–1967), who promoted the behaviorist program in psychology in the United States two years before Watson (Meyer, 1911).

Stumpf's last years in Berlin were not happy ones. During the First World War, he lost many of his American and Russian colleagues, and the student population of the Institute of Experimental Psychology was decimated. Wolfgang Köhler succeeded Stumpf as director of the Institute when he retired in 1921.

Oswald Külpe: The Würzburg School

Oswald Külpe (1862–1915) enrolled at the University of Leipzig in 1881, intending to study history and philosophy, but became interested in experimental psychology after attending some of Wundt's lectures. He moved to the University of Berlin from 1882 to 1883 to study history and to the University of Göttingen from 1883 to 1886, where he began a PhD on sensation with Müller. He returned to the University of Leipzig in 1886 and completed his degree under Wundt's supervision. From 1887 to 1894 he served as an instructor and assistant to Wundt at the Institute of Psychology. Külpe published *Outline of Psychology* in 1893, an elegant popularization of Wundtian psychology. He was appointed professor of philosophy and aesthetics at the University of Würzburg in 1894, where he developed the psychology laboratory instituted by Brentano and Stumpf; and with Karl Marbe (1869–1953), he founded the Institute of Psychology at the University of Würzburg. Külpe left Würzburg for the University of Bonn in 1909, and in 1913 he transferred to the University of Munich, where he died in 1915.

The Würzburg Institute Research at the Würzburg Institute of Psychology was directed primarily toward the experimental study of cognitive and volitional processes. The work of Külpe's associates, such as Narziss Ach (1871–1946), Karl Bühler (1879–1963), Karl Marbe, August Mayer (1874–1951), August Messer (1867–1937), Johannes Orth (1872–1949), Otto Selz (1881–1943), and Henry J. Watt (1879–1925), laid the foundations of 20th-century cognitive psychology. Distinguished students who worked at the institute included Wertheimer, Koffka, Richard Pauli (1886–1951), Charles Spearman (1863–1945), and Robert Sessions Woodworth (1869–1962). Although Külpe and his associates became engaged in a major controversy with Wundt concerning the experimental analysis of cognitive processes, the general program of the Würzburg Institute was broadly Wundtian in orientation (much more so, for example, than the rival programs of Müller at Göttingen and Stumpf at Berlin).

Experimental subjects (who included Külpe and his associates) were carefully trained in **systematic experimental self-observation** (Ach, 1905), often based upon the experimenter's active questioning or interrogation (*Ausfrage*) (Bühler, 1907/1964). Subjects were required to reflect upon and provide detailed verbal reports of any sensations or images they experienced prior to and during experimental tasks, which included free and controlled word association, choice reactions, puzzles, word problems, and psychophysical judgments. While short-lived, the work of the Würzburg psychologists marked an important transition from associationist to rule-governed theories of cognition.

Imageless Thoughts and Determining Tendencies The work of the Würzburg psychologists is important because they conducted a number of experimental studies that undermined two fundamental assumptions of associationist psychology. First, Mayer and Orth (1901), Marbe (1901), and Bühler (1907/1964) demonstrated the existence of **imageless thoughts**: “states of consciousness” and “forms of judgment” that were not associated with any sensations or images. While some of their subjects did report idiosyncratic sensations and images while forming associations or making judgments, others reported none at all. Subjects reported in experimental “protocols” that they had thoughts “without any trace of imagery.” This result, reported around the same time by Binet in Paris (Binet, 1903) and Woodworth in New York (Woodworth, 1906), eventually proved fatal to the tradition of associationist psychology from Hume to Bain in which thoughts were conceived of as images derived from sense impressions.

Second, Külpe (1912/1964), Watt (1904), and Ach (1905) demonstrated that **determining tendencies**, or “directed thoughts,” play a critical role in cognitive processing, different from (and often countervailing) associations based upon traditional principles of similarity and contiguity. They demonstrated that directive instructions override idiosyncratic associations based upon imagery when experimental instructions direct subjects to draw sentential inferences or explicate

conceptual entailments, such as stating a subordinate or superordinate category. They claimed that such instructions establish a preparatory mental schema, or “set” (*Einstellung*), that directs cognitive processing. For example, in a free association task, when asked for a response to “farmer,” subjects might say “tradesman.” When directed to respond with a coordinate category, subjects might also say “tradesman.” Yet when directed to respond with a superordinate category, subjects would generally say “occupation” (Selz, 1927/1964, p. 227). These studies suggested that traditional principles of association play only a minor role in cognitive processing and that semantic and syntactic connections between thoughts override “accidental” associations between images based upon similarity and contiguity.

As Külpe put it,

The importance of the task and its effects on the structure and course of mental events could not be explained with the tools of association psychology. Rather, Ach was able to show that even associations of considerable strength could be overcome with a counteracting task. The force with which a determining tendency acts is not only greater than the familiar reproductive tendencies, it also derives from a different source and its effectiveness is not tied to associative relations.

—(1912/1964, p. 216)

Since according to Külpe, determining tendencies play a role in all forms of thought, “the psychology of the task became an essential part of the modern investigation of thinking” (1912/1964, p. 216).

The Modern Investigation of Thinking Külpe and his colleagues not only demonstrated the limitations of associationist psychology, but also recognized the possibility of a cognitive psychology based upon the processing of thought contents in accord with rules that are *independent of image association*:

The fact that thoughts are independent of the signs in which they are expressed, and that they have peculiar and fluid interrelations, uninfluenced by the laws of the association of images, demonstrated their autonomy as a special class of conscious contents.

—(1912/1964, pp. 212–213)

Although most of the Würzburg psychologists still tended to talk in terms of *conscious* contents, they recognized that the form of consciousness involved in following a rule is quite different from the form of consciousness involved in awareness of a sensational image:

But consciousness of a rule is not thinking *of* a rule, rather it is thinking a rule or according to a rule. The object of consciousness of a rule is not the rule, but rather the state of affairs, the object, that the rule describes, on which it is used.

—(Bühler, 1907/1964, p. 163)

Bühler (1907/1964, 1908) and Selz (1922, 1927/1964) developed early forms of autonomous cognitive theory (independent of image association). Bühler was originally a student of Stumpf's at Berlin and served as an assistant to Külpe at Würzburg from 1907 to 1909. He stressed the critical role of rules in the processing of thought contents and was a pioneer of the so-called **rules and representations** approach of cognitive psychology, in which cognition is conceived of in terms of the rule-governed processing of symbolic representations (Bechtel, 1988).

Selz developed perhaps the closest approximation to contemporary cognitive theory. He received his PhD in philosophy from the University of Munich in 1909 (after having previously studied with Stumpf in Berlin). He did a postdoctoral thesis on the "laws of the ordered thought processes" under Külpe's supervision at the University of Bonn, after Külpe transferred there in 1909, and also worked with Bühler, who followed Külpe to Bonn in 1910. Selz became an instructor and junior professor at the University of Bonn from 1919 to 1921, and later professor of philosophy, psychology, and educational theory at the Commercial College of Mannheim. His theory of the role of anticipatory schema in problem solving (Selz, 1922, 1927/1964) represented the most sophisticated expression of the "modern investigation of thinking" promoted by the Würzburg psychologists (although Selz himself was never formally associated with the University of Würzburg) and anticipated later developments in cognitive psychology, particularly the computer simulation of problem solving.

The Controversy With Wundt The experimental program of the Würzburg school brought its members into conflict with Wundt, who wrote a famous critique of their experiments (Wundt, 1907). The work of the Würzburg psychologists came to be seen as a challenge to Wundt's supposed claim that "higher" cognitive processes cannot be studied experimentally, but must be explored via the naturalistic observational methods of *Völkerpsychologie*.

However, Wundt was not opposed to the Würzburg theories of cognition. His own theory of apperception precluded the equation of thoughts and sensational images, and he did not believe that human cognitive processes could be adequately explained in terms of associationist principles of similarity and contiguity. Wundt was fully supportive of the study of determining tendencies, which, he maintained, deserved "further application and development in the same direction" (1911, cited in Woodward, 1982, p. 449). The so-called **imageless-thought debate** caused a stir in America, but mainly because imageless thoughts created a problem for the form of structural psychology promoted at Cornell University by Edward B. Titchener, who interpreted Wundt's psychology in line with traditional British empiricism and associationist psychology. It was Müller, not Wundt, who defended associationist psychology against the Würzburg critiques (Müller, 1913/1964).

As noted earlier, research in Wundt's laboratory was not restricted to the study of elemental sensational processes; his students worked on apperception, memory,

and other “higher” thought processes. Wundt’s objections to the Würzburg experiments were not objections to the experimental study of “higher” thought processes per se, but were specific objections to the methodological practices of the Würzburg experimentalists, especially those of Ach and Bühler.

Wundt (1907) laid down four conditions of experimental adequacy: that the observer should be in a position to observe the process investigated and should be in a state of anticipatory attention; that experimental conditions should be varied and that experiments should be repeated. He admitted that these conditions were idealizations that are only approximated by experiments in natural science, but complained that all four conditions were violated by the Würzburg experiments. He claimed that they were “sham experiments which have the appearance of being systematic only because they take place in a psychological laboratory” and maintained that

they have no scientific value because they fall short when judged by all the criteria which distinguish the self-observations of experimental psychology from those of ordinary life.

—(1907, p. 329)

However, Wundt’s complaints about the “reprehensible method” of the Würzburg psychologists were somewhat disingenuous. As a number of later commentators noted, Wundt’s own experimental program of “trained and strained” self-observation could be faulted on his first two conditions of experimental adequacy, and the Würzburg psychologists routinely varied experimental conditions and repeated experiments (Humphrey, 1951; Woodworth, 1938). In his response to Wundt, Bühler (1908) made a reasonable case that the Würzburg experiments satisfied all four of Wundt’s conditions (a view also endorsed by Humphrey and Woodworth), at least to the degree that they were satisfied by studies in his own experimental program.

Certainly Wundt was not averse to appealing to self-observation in support of his own theories of thought processes. In taking issue with the Würzburgers’ interpretation of some of their experimental results, Wundt championed his own theoretical account of the transformation of mental configurations into sequential linguistic representations (a “higher” thought process if ever there was one), and cited as evidence his own “self-observations” of the process. Bühler (1908) expressed his surprise at Wundt’s critique because he believed that he and his colleagues had followed Wundt’s own methodological practice. Robert Woodworth, the American co-discoverer of “imageless-thoughts,” was less charitable:

He first demolishes the method of the thought experiment by his critique, and then proceeds to employ the same method himself and to reach the same results (as to

“imageless thought”) which had been reached by the Külpe school and which had seemed so objectionable.

—(1938, p. 785)

However, there was a significant difference between the experimental studies conducted by the Würzburg psychologists and those conducted in Wundt's Leipzig laboratory. Most of the experimental studies conducted in Wundt's laboratory employed objective measures of psychological processes such as reaction time, with experimental introspection normally only (and rarely) used to provide supplementary information about such processes. In contrast, the Würzburg psychologists, like Titchener's students at Cornell, made introspective reports the primary measure of experimentally investigated thought processes.

There was another perceived weakness of the Würzburg experiments. The published reports of experiments conducted in Wundt's laboratory (and other early German and American laboratories) provided the identity of the subjects engaged in experimental self-observation and described their level of training. The published experimental reports of the Würzburg psychologists failed to include such information (Bazerman, 1987; Danziger, 1990). This was no small failing for Wundt, who regularly refused to allow students to participate in experiments because they were not sufficiently trained or properly “calibrated.” Lightner Witmer, one of Wundt's American students, reported how Wundt refused to allow him to take part in reaction time experiments because he did not consider him properly calibrated as an observer: “In his opinion my sensory reaction to sound and touch was too short to be a true sensory reaction” (Witmer, letter to Boring, 1948, cited in O'Donnell, 1985, p. 35).

Other critics, such as Ebbinghaus (1902) and Müller (1911–1913), complained about the laxity of experimental controls and the danger of biasing subject responses through the use of subject interrogation, anticipating later concerns about experimenter expectancy effects (Rosenthal, 1966) and demand characteristics (Orne, 1962).

Gestalt Psychology

The best known and most influential school of psychology that developed in Germany in the years following the creation of Wundt's Leipzig program was the Gestalt school of psychology, founded by Max Wertheimer, Wolfgang Köhler, and Kurt Koffka. The basic principles of Gestalt psychology, which stressed the active and organized nature of perception, can be traced back to the work of theorists such as Kant, who claimed that the sensational elements of perception are actively organized in accord with the intuitive forms of space and time and conceptual categories such as substance and causality. The more immediate anticipators

acknowledged by the Gestalt psychologists themselves were the physicist Ernst Mach (1838–1916) and the psychologist Christian von Ehrenfels (1859–1932).

Mach did empirical research on brightness perception and movement. In *Principles of a Theory of Movement Perception* (1875/1967), he claimed that the perception of movement is determined by a sensational element additional to visual sensation (according to Mach, the inner ear is the organ that perceives movement). In *Contributions to the Analysis of Sensations* (1886), Mach claimed to have identified sensations of “space-form” and “time-form” that are independent of visual sensation.

Christian von Ehrenfels, a student of Alexius Meinong’s (1853–1920) at the Institute of Psychology at the University of Graz in Austria, introduced the notion of “form-qualities” (*Gestaltqualität*) in “On Gestalt Qualities” (1890), which he illustrated by reference to a perceived melody. The form-quality of a melody is not equivalent to the aggregate sum of its tonal elements: One may produce a different melody by rearranging of the same tonal elements in a different order, and one may produce the same melody by arranging different tonal elements (in a different key, for example) in the original order. According to Ehrenfels, the form-quality of a melody is something different from, or something “over and above,” the sum of its sensational elements. The form-quality is not determined by the sensory elements (*Fundamente*), but by the structure or pattern of their relationship (*Grundlage*): It is an emergent configuration actively created by the organizational propensities of the mind.

However, the Gestalt psychologists went beyond Mach and Ehrenfels’s stress on the emergent nature of form-qualities, which had also been recognized by Mill (as a product of cognitive inference) and Wundt (as a product of the fusion of sensational elements). They maintained not only that holistic form-qualities are underdetermined by sensational elements, but also that so-called sensational “elements” are artificial abstractions, whose nature and identity are determined by their *relational* location within a perceptual configuration (as Wundt had also claimed). The Gestalt psychologists were opposed to all forms of psychological atomism and associationist psychology: They denied that perception is a function of the combination of independent sensational elements and that perception is grounded in some form of association or cognitive inference.

The Phi Phenomenon Max Wertheimer was born in Prague. He studied law at the University of Prague but became more interested in psychology when he attended lectures by Ehrenfels. He spent a few years at the University of Berlin with Stumpf, then worked with Külpe at the University of Würzburg on his doctoral degree, which he received in 1904. From 1904 to 1910 Wertheimer held positions at the universities of Prague, Vienna, and Berlin. He taught at the University of Frankfurt from 1910 to 1916 and, after 13 years at the University of Berlin, returned as full professor in 1929. In 1933 he emigrated to the United States, where he taught at

the New School for Social Research. Wertheimer was one of a number of refugees from Nazi Germany who founded the “University in Exile” (later called the New School for Social Research) in New York, the first university in the United States devoted to adult education. He died in 1943.

According to Wertheimer’s own account (Sarris, 1997), the discovery of the phi phenomenon was the fortuitous outcome of a train journey. He was on his way to a vacation on the German Rhine during the summer of 1910. Looking out of the carriage window, he was fascinated by the telephone poles and mountains that seemed to whiz by and began to wonder about the cause of this apparent motion. He abandoned his vacation and got off the train at Frankfurt. He rented a hotel room, where he contemplated images of a moving child and horse that he produced with the aid of a stroboscope he bought at a local toy store. The next day Wertheimer visited the University of Frankfurt, where he consulted Friedrich Schumann, the former assistant to Müller and colleague of Stumpf’s, who was now working at the Institute of Psychology. Schumann had done extensive research on spatial perception (and had a PhD in physics), but could not provide an explanation of apparent motion. He offered Wertheimer the use of his laboratory, including the improved tachistoscope that he had developed. Schumann also introduced Wertheimer to two of his colleagues, Wolfgang Köhler and Kurt Koffka, assistants at the Institute of Psychology, who worked with Wertheimer and served as subjects in his motion-perception experiments. Köhler and Koffka, along with Wertheimer, are generally recognized as the joint founders of Gestalt psychology, or the **Berlin (or Berlin-Frankfurt) school of Gestalt psychology**.

In Wertheimer’s original experiments, he projected flashes of light successively through two horizontal slits onto a screen. At a projection rate of about 50–60 milliseconds, a single line of light appeared to move from one position to another. At higher rates, two lines of light appeared simultaneously, and at lower rates they appeared individually in succession. Wertheimer called the impression of motion generated by alternative illumination at the middle projection range the **phi phenomenon**. He reported the results of these experiments in his 1912 paper “Experimental Studies in the Visual Perception of Motion.” Vittorio Benussi (1878–1927), a student of Meinong’s at the University of Graz, reported a similar phenomenon with respect to the sense of touch: When points on the skin are stimulated in rapid succession, they produce a subjective impression of apparent motion, as if the tactile stimulus moves in an arc through space (Benussi, 1914).

On the basis of these experiments, Wertheimer concluded, somewhat paradoxically, that apparent motion is *perceived*. What he meant was that our impression of both apparent and real motion is not the product of association or cognitive inference made on the basis of discrete sensational elements, as Berkeley, Mill, and Helmholtz had maintained. Wertheimer rejected the prevalent explanation of the perception of motion as an inference based upon eye movements. He demonstrated that subjects still perceive apparent motion when they are asked to visually

fixate on a single central point and when the cycle of illumination is less than the minimal reaction time required for eye movements (Lowry, 1982).

Relational Elements Wertheimer denied that visual stimuli on the retina give rise to punctiform visual sensations in the brain, which form the basis of a distinct process of association or inferential judgment. Rather, he maintained that retinal input is processed directly by the brain to generate the perception of motion, via what he called “a kind of physiological short circuit” (*Kurzschluss*) (1912/1925, p. 88). According to Wertheimer, excitations in the brain become integrated into a kind of “physiological whole-process” (*Gesamtprozess*) (1912/1925, p. 92).

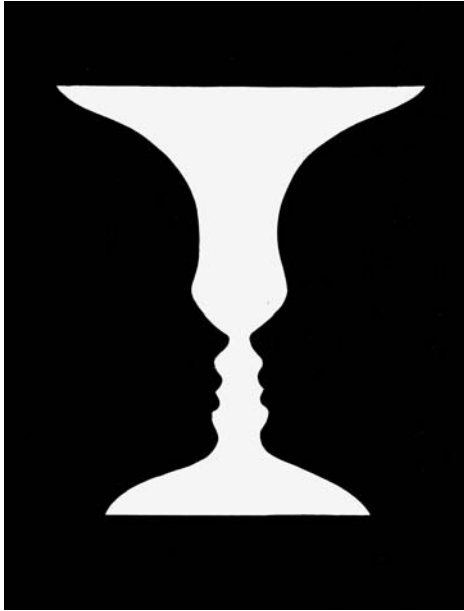
This last point is important to stress, because it grounded the central thesis of the Gestalt psychologists: the denial of the existence of atomistic sensational elements (Ash, 1995). According to Wertheimer (and Köhler and Koffka), sensational elements are not autonomously determined by the external world and our sensory receptors, and perception is not a product of “senseless additive combining” (Lowry, 1982, p. 186). Although we can conceptually distinguish elements of perception, the nature and identity of such elements is relationally determined by their location within perceptual configurations or structures. One configuration of sensory input will generate a perception analyzable in terms of one set of elements; another configuration of the *same* sensory input will generate a quite different perception, analyzable in terms of a different set of elements. As Wertheimer put it,

There are wholes, the behavior of which is not determined by that of their individual elements, but where the part-processes are themselves determined by the intrinsic nature of the whole.

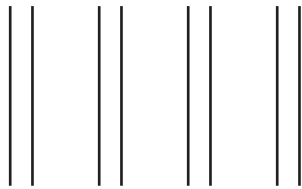
—(1925/1938, p. 2)

Probably the best phenomenological illustration of this claim is the old woman/young woman ambiguous figure. When the visual input is configured one way, a particular point on the visual image represents the end of the old woman’s nose; when the visual input is configured another way, the same point represents the end of the young woman’s chin. The identity of the point *as* the end of the old woman’s nose or the young woman’s chin is determined by its relational location within the perceptual configuration. As Wertheimer put it, “‘Elements’ . . . are determined as parts by the intrinsic conditions of their wholes and are to be understood ‘as parts’ relative to such wholes” (1922/1938, p. 14).

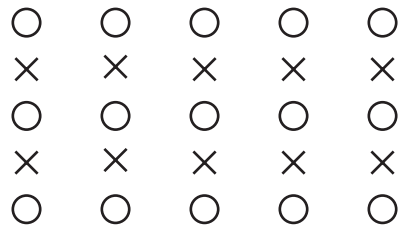
Good Form Wertheimer, Köhler, and Koffka founded the journal *Psychological Research* (*Psychologische Forschung*) in 1921 to publish experimental reports and theoretical articles relating to the new Gestalt psychology. According to the Gestalt psychologists, the elements of perception are actively organized into wholes according to various principles, such as continuity (elements grouped in lines tend



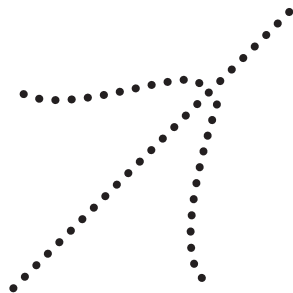
Ambiguous figures: vase/faces and old woman/young woman.



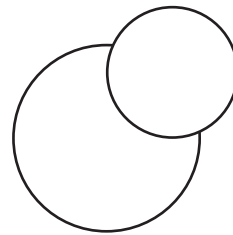
Proximity



Similarity



Good continuation



Closure

Good form: Gestalt principles of proximity, similarity, good continuation, and closure.

to be perceived as continuous), inclusiveness (elements tend to be perceived as the largest possible figure), similarity (similar elements tend to be grouped together), proximity (temporally or spatially adjacent elements tend to be grouped together), and closure (incomplete configurations tend to be completed). According to the **law of Prägnanz**, or law of good form, sensational input will be organized into a form that is as concise, ordered, and proportioned as is possible given the conditions.

Gestalt psychological theory worked best with respect to the explanation of perceptual illusions. These created a problem for standard empiricist accounts of perception, but were readily accommodated by Gestalt accounts in terms of alternative configurations of stimulus input, as in Koffka's account of our perceptual responses to the Necker cube (1935). Analogously, different perceptual responses to ambiguous figures, such as the familiar vase/faces and old woman/young woman figures, could be readily explained in terms of different perceptual configurations of sensory input, in terms of different figure-ground configurations, for example (Rubin, 1915/1921).

Koffka and Köhler Kurt Koffka was born in Berlin and took his PhD with Stumpf in 1909. He served as an assistant at the universities of Würzburg and Frankfurt from 1909 to 1910 and as lecturer and later professor at the University of Giessen from 1911 to 1924. He visited the United States in 1924, where he served as a visiting professor at Cornell University and the University of Wisconsin. Eventually he accepted a permanent position at Smith College in Massachusetts. His 1922 *Psychological Bulletin* article introduced Gestalt psychology to many Americans, although it focused almost exclusively on perception. His 1935 book *Principles of Gestalt Psychology* was broader in scope, but the difficulty of the work limited its influence. Koffka also extended Gestalt principles to child development in *The Growth of the Mind: An Introduction to Child Psychology* in 1924.

Wolfgang Köhler was perhaps the most effective advocate of Gestalt psychology and became its informal spokesperson. He was born in Estonia and originally trained as a physicist with Max Planck (1858–1947). He took his PhD with Stumpf in 1909 and accepted a position at the University of Frankfurt that same year. In 1913 the Prussian Academy of Sciences invited him to serve as director of its anthropoid station in Tenerife. During his years there Köhler did pioneering research with primates, documented in *The Mentality of Apes* (1917), in which he extended the principles of Gestalt psychology to animal learning. In one series of studies, chimpanzees demonstrated their ability to stack boxes to reach hanging bananas and to manipulate sticks of different lengths to access bananas placed outside their enclosure. Köhler claimed that the chimpanzees' success in these tasks was a product of **insight learning**, a creative form of learning that involves perceptual restructuring of the problem situation, in contrast to the mechanical forms of "trial-and-error" learning that comparative and animal psychologists such as Morgan and Thorndike attributed to animals. In opposition to Morgan and later behaviorists, Köhler maintained



Insight learning: chimpanzee stacking boxes.

that chimpanzees engage in means-ends reasoning and that many animals are capable of perceiving relations (including animals of lower intelligence such as chickens). In what Köhler called **transposition learning**, animals respond to similarities in relationships between stimuli rather than similarities between stimuli themselves. For example, chimpanzees trained to choose a 15-cm over a 10-cm disk, and subsequently presented with 15-cm and 20-cm disks, choose the larger 20-cm disk: they make *relational* rather than stimulus-specific responses.

It has been suggested that Köhler's business in Tenerife was not entirely academic. He may have been involved in espionage during the First World War, sending out radio reports on the movement of British shipping (Ley, 1990), although the suggestion has been disputed by Harris (1991). Köhler returned to take up a position at the University of Göttingen in 1921. The following year he succeeded Stumpf to the chair in philosophy at the University of Berlin, with directorship of the Institute of Psychology, now one of the premier positions in the German psychological academy. Köhler visited the United States in the 1930s, where he served as a visiting professor at Clark, Harvard, and Chicago. His influential statement of the general principles of Gestalt psychology was published as *Gestalt Psychology* in 1929.

From his position of academic authority, Köhler bravely criticized the Nazis and defended the Jews (Henle, 1978a). He expected to be arrested at any time, although he never was. He resigned from the Berlin Institute of Psychology in 1934, but the authorities would not accept his resignation. Köhler visited Harvard in 1934, where he gave the William James Memorial Lecture and a series of public lectures and graduate seminars. He was considered for an appointment at Harvard, but Edwin G. Boring, the chair of the psychology department, opposed his candidacy. Boring wanted an accredited experimentalist and successfully lobbied in favor of the neurophysiologist Karl Lashley (1890–1958).

Köhler emigrated to the United States in 1935, when things became intolerable in Nazi Berlin (he refused to take the Nazi loyalty oath, and the authorities finally accepted his resignation). He secured a full-time position at Swarthmore College in Pennsylvania, where he remained until he retired in 1958. Köhler continued writing and conducting research at Princeton, Dartmouth, and MIT until he died in 1967. He published *The Place of Value in a World of Facts* in 1938 and

Dynamics in Psychology in 1940. Köhler received an APA Distinguished Scientific Contribution Award in 1956 and was elected president of the APA in 1959. Some found him aloof and ill-humored as a person, but not the graduate students at Clark University, whom he taught to tango during his visit there.

Gestalt Psychology and Field Theory Köhler also developed the fundamental neurophysiological theory of Gestalt psychology (modeled on recent advances in physical field theory), which received its formal statement in *Static and Stationary Physical Gestalts* in 1920. According to Köhler, the basic principles of Gestalt psychology are a function of the “inherent dynamical properties of the brain.” Force fields in the brain tend to seek equilibrium and remain in stationary equilibrium until some external force disturbs them. When so disturbed, force fields become dynamic and strive to regain equilibrium.

Köhler reductively explained perception in terms of the principle of **psychoneural isomorphism**: Configured perceptual fields are structurally identical to configured neural fields, of which they are products. If sensational input generates a balanced neural field, perceptual output will correspond to sensational input. If sensational input generates an unbalanced neural field, the field will reconfigure itself and perceptual output will not correspond to sensational input: The neural elements corresponding to the sensory input will be *transformed* in the reconfiguration of the neural field to attain equilibrium. The law of Prägnanz was held to be a consequence of the reconfiguration of neural fields to attain equilibrium. Thus, in the case of space perception, for example, Köhler held that “experienced order in space is always structurally identical with functional order in the distribution of underlying brain processes” (1929, p. 61).

This neural theory was entirely speculative, and Köhler freely admitted that Gestalt psychologists were “not in the position to derive the respective physiological and phenomenal characteristics [of psychological processes] in individual cases” (1920, p. 259). The rare experimental studies of neural field theory proved negative. Lashley manipulated electrical fields on the surface of the brains of monkeys, but with no discernible effect on their performance on visual discrimination tasks (Lashley, Chow, & Semmes, 1951).

Wertheimer, Köhler, and Koffka extended the basic principles of Gestalt psychology beyond perception to provide explanations of reflexive behavior, the ego, adjustment, will, memory, thought, personality, and the social world. Karl Duncker (1903–1940) and Wertheimer developed Gestalt principles to provide a general theory of problem solving in *Psychology of Thinking* (Duncker, 1935/1945) and *Productive Thinking* (Wertheimer, 1945). Köhler extended Gestalt principles to the study of memory (Köhler & von Restorff, 1933) and Koffka to the study of child development (1924). Kurt Gottschaldt (1902–1901), who studied with Wertheimer and Köhler, taught at the University of Berlin from 1935 to 1962 and extended Gestalt principles to developmental psychology and personality theory

(he was the only member of the Berlin school of Gestalt psychology who did not emigrate to America during the Nazi era).

Koffka articulated the basic justification for this extension in *Principles of Gestalt Psychology* (1935). He claimed that the law of Prägnanz applied to all psychological processes, since all psychological processes are a function of dynamic forces in the brain. According to Koffka, all psychological processes involve some form of tension displacement: Dynamic forces are brought into balance as the neural system reduces the tension caused by disequilibrium. On the psychological level, the tension created by some perceived problem, for example, motivates our attempts to resolve it, which involves cognitive scanning and imaginative rehearsal, as in the case of Wertheimer's account of productive thinking, Köhler's account of insight learning in primates, and Koffka's account of memory.

A good illustration of this principle was the study that Bluma Zeigarnik conducted on the memory of waiters. Zeigarnik studied with Kurt Lewin, who was a colleague of Köhler's and Koffka's at the Berlin Institute of Psychology in the 1920s. Lewin noted that Berlin waiters tended to remember the details of a client's bill while it remained unpaid, but immediately forgot them once the bill was paid. Zeigarnik explored Lewin's suggestion that the tension created by the lack of cognitive closure facilitates recall. Once payment is received, closure is achieved, dissipating tension and erasing memory. Zeigarnik found that interrupted and unfinished tasks tend to be remembered better and quicker than uninterrupted and completed tasks (Zeigarnik, 1927).

The Support for Gestalt Psychology Many of the theoretical explanations offered by the Gestalt psychologists, including Wertheimer's original explanation of the phi phenomenon, were highly speculative and undersupported by experimental data. Wertheimer's experiments demonstrated the inadequacy of a specific account of the perception of motion based upon eye movements, but not the general inadequacy of accounts based upon association or cognitive inference (Lowry, 1982).

Müller, Bühler, Spearman, and members of the Leipzig school of Gestalt psychology were critical of the work of the Berlin Gestalt psychologists. Members of the **Graz school of Gestalt psychology**, located at the Institute of Psychology at the University of Graz, advanced an alternative account of the production of perceptual configurations. Stephan Witasek (1870–1915), a former student of Meinong's, claimed that different perceptual configurations are psychological products of preexisting sensational elements (Witasek, 1908). He and his associates Vittorio Benussi (1878–1927) and Alois Höfler (1853–1922) maintained that this explained how *different* perceptual configurations could be generated from the *same* sensory elements:

Between the sensory impressions, which remain constant, and the perception of figures, which may differ from each other, an event X must take place, which,

depending on the form it takes, will lead to the perception of totally different objects from the same constant sensory stimulation.

—(Benussi, 1914, p. 400, cited in Fabian, 1997, p. 254)

The members of the Graz school agreed that perceptual wholes have properties that are not an aggregative function of the properties of their sensory elements and that the same sensory elements can result in different perceptual configurations. However, they denied that the creative psychological processes involved in perception transform the *identity* of the original sensory elements: They denied that the identity of sensory elements is determined by their relation to other elements in a configured whole.

On the face of it, this distinctive thesis of the Berlin school of Gestalt psychology was hard to defend. Recalling Ehrenfels's example of a melody, Wertheimer claimed that the identity of a tone derives from its role in a particular melody: "The flesh and blood of a tone depends from the start upon its role in the melody" (1925/1938, p. 5). Yet tones appear to exist and be identifiable *independently* of the melodies in which they often occur. Indeed, this is presupposed by the common Gestalt psychological claim, originally advanced by Ehrenfels, that the *same* tonal elements of one melody can be rearranged to form a *different* melody: The identity of the tonal elements remains the same, but the melody changes with the different configuration of the elements. Ironically, this atomistic assumption about the identity of sensory elements appears central to some of the explanations the Berlin Gestalt psychologists offered for a variety of perceptual phenomena, such as perceptual constancy, in which perceived objects remain the same shape and size despite changes in elemental sensory inputs (supposedly because of invariance in figure-ground ratios in perceptual and brain fields).

Even the old woman/young woman ambiguous figure doubtfully serves as an illustration of the transformation of sensory elements in perceptual configurations. Edgar Rubin, the former student of Müller's who created many of the ambiguous figures of Gestalt psychology, thought it demonstrated only that the same sensory elements could be "completed" in different ways to generate different perceptions (Rubin, 1915/1921). David Katz, another student of Müller's, probably provided the best empirical support for the claims about the relational identity of sensational elements championed by the Berlin school. Katz's (1911) research on "phenomenal modes of color" indicated that the phenomenal appearance of colors depends upon the context in which they are viewed, as "surface," "volumic," or "film" colors (Lowry, 1982).

The Legacy of Gestalt Psychology The limited empirical support for Gestalt theories of perception cast doubt upon their extension to other psychological domains, and continued interest in Gestalt psychology has tended to remain restricted to perception. The theory of psychological isomorphism, which grounded the

explanatory reduction of psychological configurations to neural configurations, was undermined by the failure of neural field theory. Köhler's own neurophysiological research at MIT did not provide any empirical support for the theory.

The Gestalt psychologists were dismissive of alternative explanations of the various perceptual, cognitive, and social psychological processes covered by their theories. They promoted Gestalt psychology with evangelical zeal, behaving like "intellectual missionaries, spreading a new gospel" (Sokal, 1984, p. 1257), and located the principles of Gestalt psychology within a general holistic philosophical worldview. These features partially account for the limited impact of Gestalt psychology on the development of American psychology.

While the Gestalt psychologists were accorded due respect by the American psychological community, they attracted few American disciples to continue their general theoretical program. One reason for their lack of impact was the fact that Wertheimer and Koffka died fairly young, and they all held positions at institutions without graduate programs in psychology (Wertheimer at the New School, Koffka at Smith College, and Köhler at Swarthmore College).

Although Gestalt psychology was opposed to the atomism of traditional associationist psychology and behaviorism, it was as thoroughly mechanistic as both. Some have come to associate Gestalt psychology with an emphasis on human agency and creativity and a holistic rejection of mechanistic and reductive approaches to the explanation of human behavior. "Gestalt therapy" has also been championed as a form of "self-discovery," as opposed to more directive forms of therapy (Perls, Hefferline, & Goodman, 1951). However, there is little support for such characterizations in the work of Wertheimer, Köhler, and Koffka, who all insisted that psychological processes are wholly determined by automatic neural processes. Although they made some theoretical contributions to psychotherapy (Knapp, 1986), Köhler explicitly repudiated the form of "Gestalt therapy" promoted by Fritz Perls (1893–1970) and his colleagues (Henle, 1978b).

APPLIED PSYCHOLOGY IN GERMANY

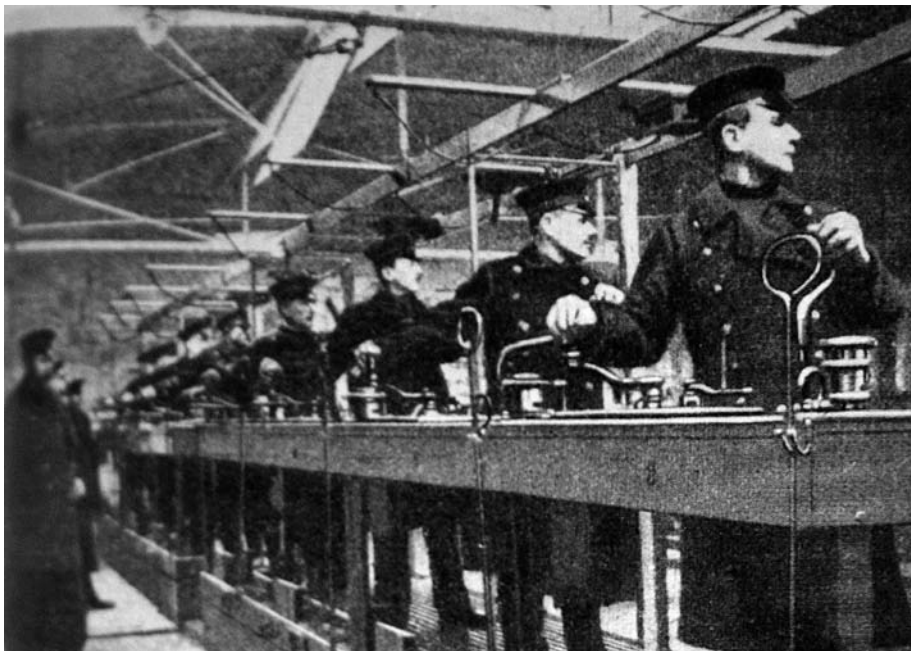
Despite their theoretical differences (and personal animosity), Wundt and Stumpf saw psychology as a natural development of philosophy. However, the majority of German philosophers and later psychologists did not share this view. Wilhelm Dilthey, Stumpf's philosophical colleague at the University of Berlin, claimed that natural scientific methods are inappropriate for the study of psychological states and processes and engaged in critical debates with Wundt, Stumpf, and Ebbinghaus about the legitimacy of experimental psychology.

Opposition to the appointment of psychologists to philosophy chairs in Germany reached such a pitch that in 1913 Wilhelm Windelband (1848–1915) was able to organize a petition by 107 philosophers in Germany, Austria, and

Switzerland, who protested against “the filling of chairs in philosophy with representatives of experimental psychology” (quoted in Ash, 1980, p. 407). Opposition from philosophers threatened to overwhelm psychology in Germany (Metzger, 1965), forcing Wundt to defend both experimental psychology and its integral relation to philosophy in *Psychology Struggling for Survival* (1913).

After the First World War, experimental psychologists were no longer appointed to chairs in philosophy, and traditional philosophers recaptured the chairs at the universities of Bonn and Wroclaw. New positions were introduced for psychologists during the 1920s and 1930s, but these were almost exclusively at technical universities and institutes (Kusch, 1995). German psychologists eventually followed the lead of their philosophical critics and organized their own petition in 1931 for the establishment of separate chairs of psychology at leading German universities.

Wundt had rejected what he called the “American model” of separate psychology departments and chairs, which he feared would lead to an overemphasis on applied over theoretical and experimental psychology. Yet this is precisely what happened in Germany in the years after Wundt’s death. During the 1920s psychology in Germany became increasingly applied, and by 1925 publications in applied psychology outnumbered those in “pure” or general psychology by two to one (Osier & Wozniak, 1984, cited in Kusch, 1995, p. 124).



Psychotechnics: group training facility for streetcar drivers in Berlin.

Karl Marbe, who had studied with Wundt and Külpe, turned to the psychology of advertising and studied the psychology of accidents and industrial damage. He pioneered the development of **psychotechnics** by creating aptitude tests for train conductors, insurance agents, prison guards, dentists, and surgeons and played a major role in instituting the subdisciplines of school and forensic psychology.

While serving as an assistant at Wundt's Leipzig Institute, Ernst Meumann (1862–1915) directed work on educational psychology, which he called “experimental pedagogy.” A progressive social and school reformer like John Dewey (1859–1952) and G. Stanley Hall in the United States, Meumann held posts in psychology and education at various universities, until he settled as director of the philosophical seminar and psychology laboratory at the Hamburg Colonial Institute. Wundt thought him an outstanding student, and they remained close

friends, although Wundt withdrew his original support for Meumann's research, because of the limited number of publications that came out of it. Meumann's *Introductory Lectures on Experimental Pedagogy and Its Psychological Basis* (1907) became a required text for generations of education students in Germany and was well received in North and South America and Russia. Meumann founded *The Archives of Psychology* in 1903 and the *Journal of Experimental Education* (later the *Journal of Educational Psychology*) in 1911. His studies of what became known as “social facilitation” in the classroom were later developed by Floyd Allport (1890–1978), one of the founders of American social psychology (Allport, 1920, 1924).

William Stern (1871–1938), a former student of Ebbinghaus's, took over from Meumann at the Hamburg Colonial Institute in 1916 (after 19 years at the University of Breslau). He became director of the New Psychological Institute (which he helped to found) in 1919. Stern was committed to the application of psychological knowledge to industry, commerce, and education and founded the *Journal of*



Farm Kitchen: Picture used in William Stern's eyewitness research.

Applied Psychology in 1908 (with Otto Lipmann). He introduced the term **differential psychology** to characterize the study of individual differences, originally in the context of his own pioneering studies of eyewitness testimony (Stern, 1902). He also introduced the notion of a **mental quotient**, defined as a child's mental age (which he held to be determined by performance on the Binet-Simon test of intelligence) divided by its chronological age (Stern, 1914). Stern also pioneered the development of personality psychology and had a special influence on Gordon Allport (1897–1967), the Harvard social psychologist and founder of personality theory in America (Allport, 1937). With his wife Clara, Stern published some classic studies of child language and memory. His work was well received by American educators and psychologists, and he accepted a position at Duke University in 1934, where he died suddenly in 1938.

Karl Bühler, another of Külpe's students, founded the Psychological Institute at the University of Vienna (sometimes known as the Vienna School of Psychology) in 1922. It became known for its tradition of rigorous experimental research and application, particularly through its productive association with the Institute of Education. Karl was joined by his wife Charlotte Bühler (1893–1974) in 1923, the first female instructor in Germany at Dresden Technical University. Charlotte, who was known for her original research on teenagers (Bühler, 1918), established the Institute of Child Psychology. The Böhlers pioneered the use of naturalistic developmental studies at the Vienna Reception Center for Children, a children's shelter now known as the Charlotte Bühler home. Charlotte visited the United States on a Rockefeller Foundation Fellowship in 1924–1925 and managed to secure 10 years of funding for the Vienna Institute from the Foundation (Rollett, 1997).

Famous students of the Institute included Paul Lazarfeld (1901–1976), the social psychologist, methodologist, and statistician; Else Frenkel-Brunswik (1908–1958), wife of Egon Brunswik (1903–1955) and one of the coauthors of *The Authoritarian Personality* (Adorno et al., 1950); and Karl Popper (1902–1994), the future philosopher of science, whose PhD thesis on "The Genetic Theory of Intelligence" was examined by Bühler and Moritz Schlick (1882–1936), one of the founders of logical positivism. Karl Bühler was a visiting professor at Johns Hopkins and Stanford and received an invitation to join the faculty at Harvard in 1930, but unfortunately turned it down. When the Nazis annexed Austria in 1938, he was demoted, then discharged and imprisoned. The new director Gunther Ipsen (1889–1984) introduced "racial psychology" the following year.

By this time psychology in Germany, now increasingly applied, had been appropriated by the military. Many of the leading German psychologists had already fled to the United States (including Wertheimer, Köhler and Koffka, Lazarfeld, Frenkel-Brunswik, and William and Clara Stern), and others suffered at the hands of the Nazis. Otto Selz, Külpe's student and anticipator of 20th-century cognitive psychology, perished in the concentration camp at Auschwitz.

DISCUSSION QUESTIONS

1. Wundt believed that scientific psychology should be based upon the experimental methods but not the explanatory concepts of physiology. Sechenov believed that scientific psychology should be based upon the experimental methods and the explanatory concepts of physiology. Who do you think was right? Why?
2. Wundt characterized his configurational psychology as active and voluntary, in supposed contrast to passive associationist psychology. In a mechanistic psychology, is there any difference between active as opposed to passive perception and cognition?
3. According to Wundt's principle of psychical relations, psychological "elements" are relational in nature. Wundt's illustrative example was words, which he claimed have meaning only in the sentences in which they are configured. Was this a good example of a relational psychological element?
4. Is intentionality the mark of the mental, as Brentano maintained? Can you think of any mental states that are not intentional?
5. Did Wertheimer's experiments on the phi phenomenon demonstrate that perceptual "elements" are relational in nature? Did any Gestalt studies demonstrate this?

GLOSSARY

accommodation In Herbart's theory of learning, the process by which an apperceptive mass is adjusted to incorporate a new representational element.

act psychology Term used to characterize Brentano's psychology concerned with mental acts, as opposed to Wundt's supposed psychology of mental contents.

apperception In Herbart, focused attention. In Wundt, the creative and selective attentional process responsible for the configuration of conscious mental states.

apperceptive mass According to Herbart, the constellation of connected elementary mental representations that constitute the current object of apperception or focused attention.

assimilation In Herbart's theory of learning, the process by which a new representational element is integrated with an apperceptive mass.

- Berlin (or Berlin-Frankfurt) school of Gestalt psychology** Form of Gestalt psychology associated with Max Wertheimer, Wolfgang Köhler, and Kurt Koffka, who maintained that the nature and identity of perceptual “elements” is determined relationally by their position within perceptual configurations or structures.
- complication experiment** Form of reaction-time experiment in which the time taken for components of a complex task is calculated by subtraction of the measured time taken for other components of the task.
- determining tendency** Structure of task that determines cognitive processing independently of image association, experimentally identified by members of the Würzburg school.
- differential psychology** Term introduced by William Stern to describe the study of individual differences.
- experimental self-observation** Form of self-observation in which trained subjects provide concurrent commentaries on their conscious experience under rigorously controlled experimental conditions.
- Graz school of Gestalt psychology** Form of Gestalt psychology associated with psychologists at the University of Graz, who maintained that perceptual configurations are psychological products of preexisting sensational elements.
- homeostatic motivation** Theory of motivation according to which organisms are motivated to eliminate states of disequilibrium that are experienced as unpleasant or painful.
- imageless thought** Instances of thought not accompanied by sensations or images, experimentally identified by members of the Würzburg school.
- imageless-thought debate** The debate about the existence of imageless thoughts between members of the Würzburg school and Edward B. Titchener, the representative of structural psychology in the United States.
- immediate experience** The kind of experience that is not subject to theoretical interpretation. According to Wundt, the subject matter of psychology.
- insight learning** According to Köhler, a creative form of learning that involves perceptual restructuring of the problem situation.
- law of Prägnanz** In Gestalt psychology, law of good form.
- mediate experience** The kind of experience that is theoretically interpreted. According to Wundt, the subject matter of natural science.
- mental chronometry** Use of complication experiments to estimate the duration of postulated mental processes such as discrimination and choice.
- mental quotient** As defined by William Stern, a child’s mental age divided by its chronological age.

modern investigation of thinking Form of cognitive psychology based upon the rule-governed processing of thought contents developed by members of the Würzburg school.

phi phenomenon Perception of apparent motion generated by projected light.

propaedeutic science A foundational science.

psychical resultants, principle of Formulated by Wundt, assertion that the attributes of psychological configurations that are the product of apperception are distinct from the mere aggregation of the attributes of the elements from which they are configured. Also known as the principle of creative synthesis.

psychical relations, principle of Formulated by Wundt, assertion that the nature and identity of the elements of psychological configurations is determined by their relational location within psychological configurations.

psychology of contents Term used to characterize Wundt's experimental analysis of the elements of consciousness, in contrast to Brentano's act psychology.

psychoneural isomorphism Theory that configured perceptual fields are the product of structurally identical neural fields.

psychotechnics Early German name for aptitude testing.

rules and representations The approach in cognitive psychology in which cognition is conceived of in terms of the rule-governed processing of symbolic representations.

systematic experimental self-observation Method of self-observation associated with the Würzburg school, in which untrained subjects were required to produce detailed verbal reports of sensations or images associated with experimental tasks, often based upon active questioning or interrogation by experimenters.

threshold of consciousness In Herbart, the level below which unconscious ideas are repressed.

transposition learning Form of learning identified by Köhler in which animals respond to similarities in relationships between stimuli rather than to similarities between stimuli themselves.

tri-dimensional theory of feeling Wundt's theory that feelings vary along three dimensions: pleasant versus unpleasant, high versus low arousal, and concentrated versus relaxed attention.

Völkerpsychologie Comparative-historical form of "folk" (or "cultural" or "social") psychology that Wundt considered to be an important supplement to experimental psychology.

voluntaristic psychology Wundt's theoretical psychology, so-called because of his emphasis on the voluntary, selective, and creative nature of the central control process of apperception.

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Psychology in America: The Early Years

PSYCHOLOGY FIRST BECAME AN ACADEMIC SCIENTIFIC DISCIPLINE IN Germany because Germany developed the modern university system that enabled Wundt and his colleagues to create an experimental psychology grounded upon the earlier success of experimental physiology. However, psychology developed faster institutionally in America than it did in Germany (Ash, 1980; Danziger, 1979). By the end of the 19th century, American psychology had an active professional organization, the American Psychological Association, founded in 1892; journals devoted to general, experimental and applied psychology, such as the *American Journal of Psychology*, *Pedagogical Seminary*, *Psychological Review*, *Psychological Index*, and *Psychological Monographs*; and a substantial academic presence within the American university system. By 1892, America had more and better laboratories than Germany. Formal research laboratories were established at the University of Chicago, Clark University, Columbia University, Cornell, Harvard, Pennsylvania, Wisconsin, and Yale; and demonstration facilities for teaching and training were available at Brown, Catholic University of America, Indiana, Kansas, Michigan, Nebraska, Illinois, Iowa, and Wellesley (Hale, 1980; O'Donnell, 1985). By 1904, there were 49 laboratories, 169 members of the APA, and 62 institutions offering three or more courses in psychology; in addition, psychology ranked fourth in the sciences with respect to the number of PhDs awarded (Camfield, 1973; Miner, 1904). By 1913, the year that Watson published his behaviorist manifesto "Psychology as the Behaviorist Views It," America had surpassed Germany in research publications (Cattell, 1917).

Much the same was true of the academic institutionalization of other social sciences in the early 20th century. Émile Durkheim (1858–1917) introduced sociology in France at the end of the 19th century, but it made slow progress, and by 1930 there were only three chairs in sociology. Yet by 1910 there were about 50 full-time professors of sociology in America, and the subject was taught at around 400 colleges (Smith, 1997).

In contrast to the often reactionary response to attempts to introduce psychology in Britain and Europe, in America it was generally accepted by the public,